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DATA VISUALIZATION

PRACTICAL - 12

AIM: Sharing, Publishing, and Presenting Data Visualizations.

a) Export dashboards as PDF/Image
b) Use presentation mode in Tableau
c) Publish dashboards to Tableau Public
d) Share dashboards via Tableau Server/Online
e) Enable user interaction on shared dashboards
f) Manage permissions and access control
g) Evaluate and optimize sharing strategies

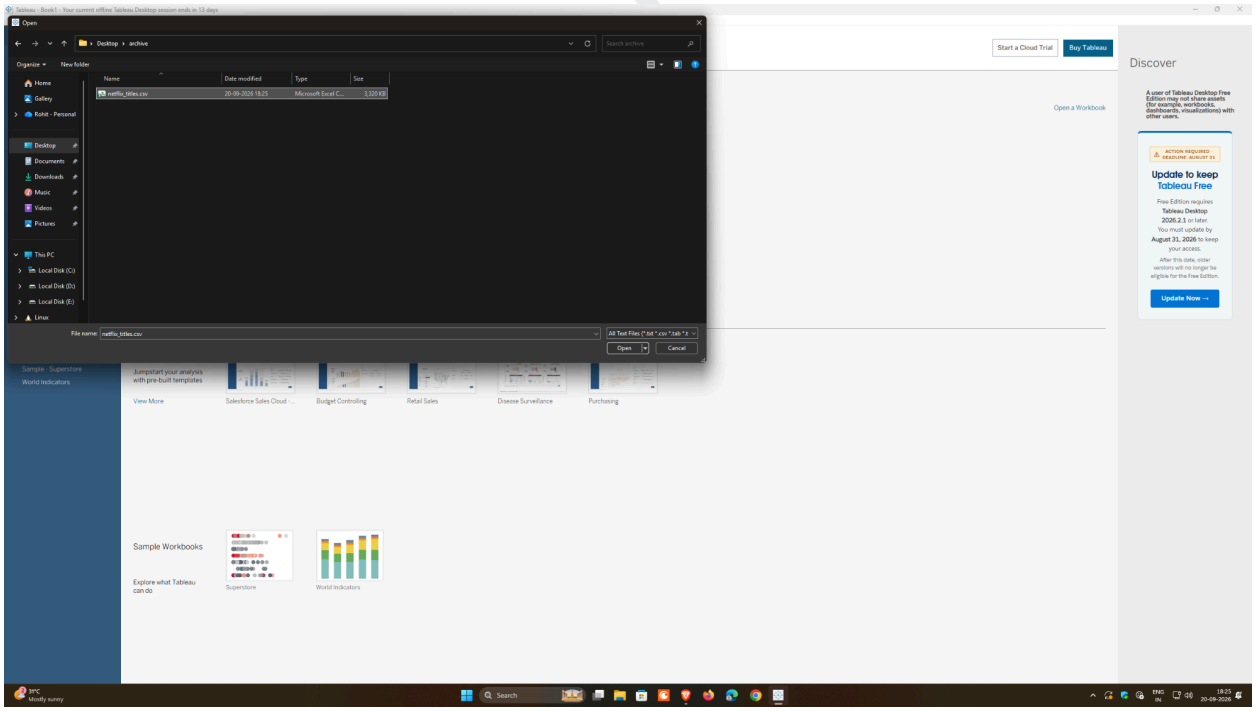
Dataset: [Netflix Movies and TV Shows]

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A: Export Dashboards as PDF/Image

Step 1: Build Two Base Visualizations

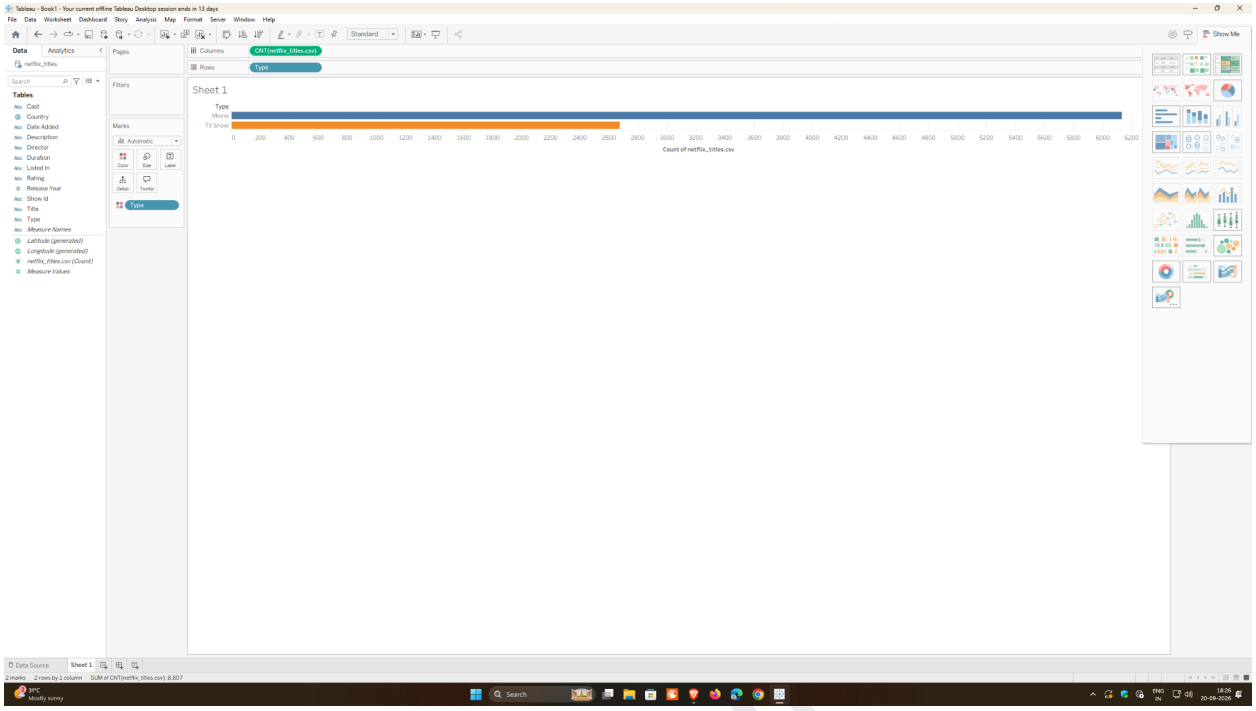
Connect **netflix_titles.csv** as a Text File, then drag **Count of netflix_titles/Number of Records** → **Columns** and **Type** → **Color** to create a horizontal **Movies vs. TV Shows** bar chart.



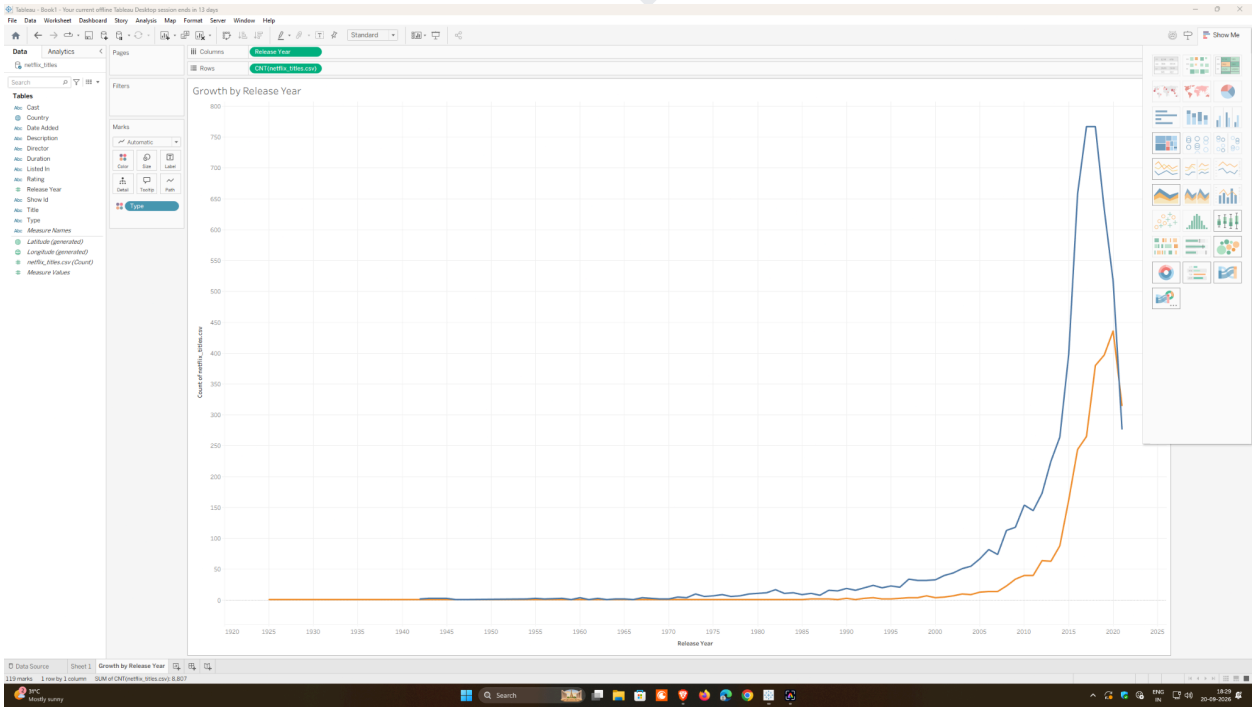
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Sheet 2: Release Year Growth (Line Chart)
Create **Growth by Release Year**, then drag **release_year** → **Columns (Continuous)**, **Count of netflix_titles** → **Rows**, and **Type** → **Color**.

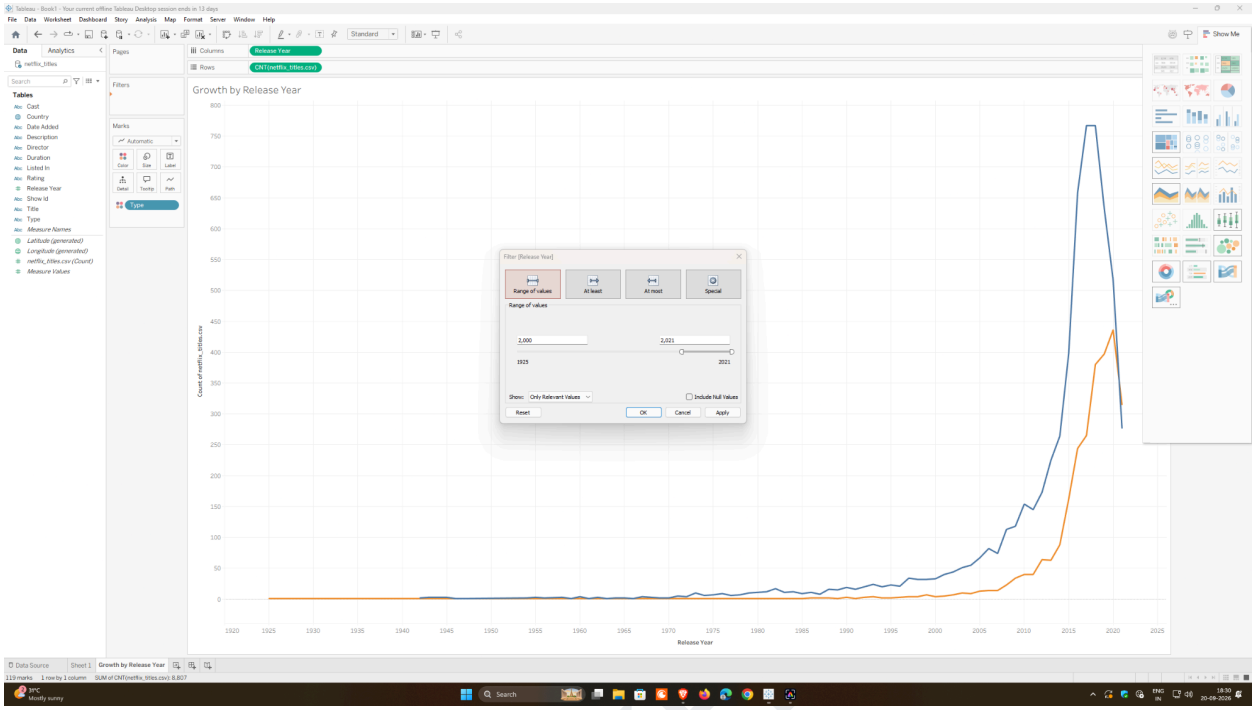


Set **release_year** filter to **2000–2021 (Range of Values)** to remove sparse historical data and display the **dual-line content volume trend**.

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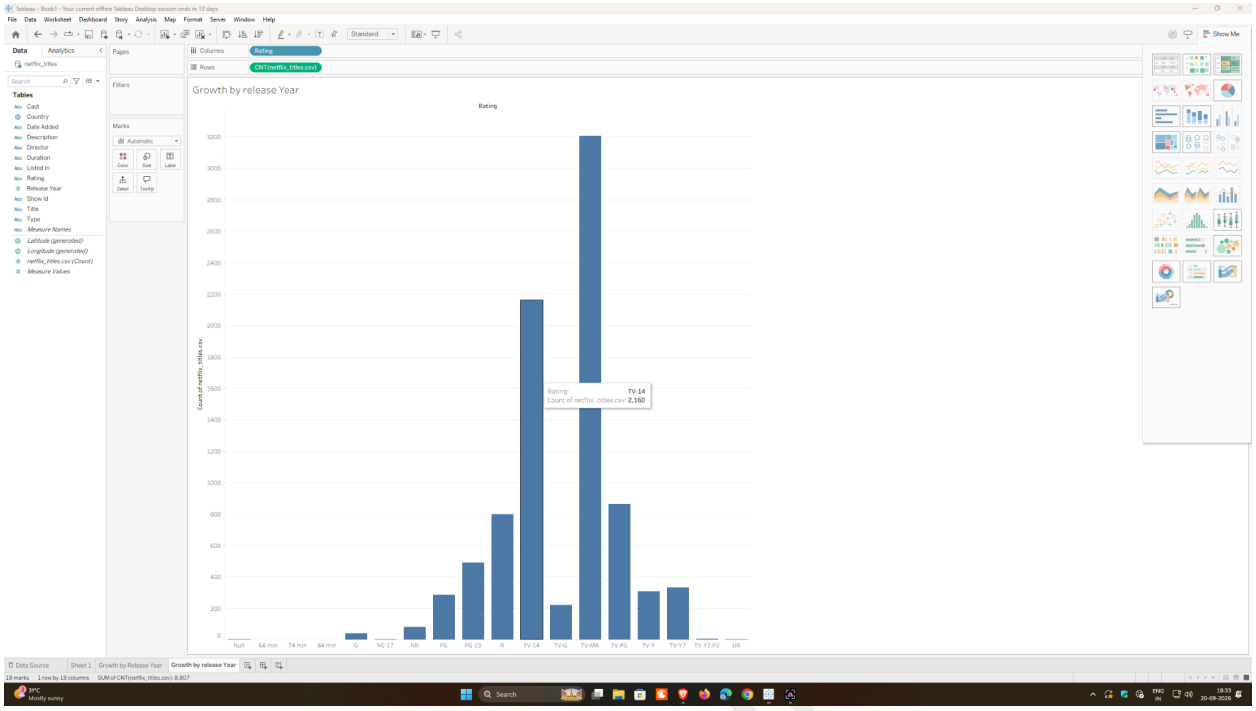
Sheet 3: Top 10 Ratings
Create **Growth by Release Year**, then drag **Rating** → **Rows** and **Count of netflix_titles** → **Columns**.

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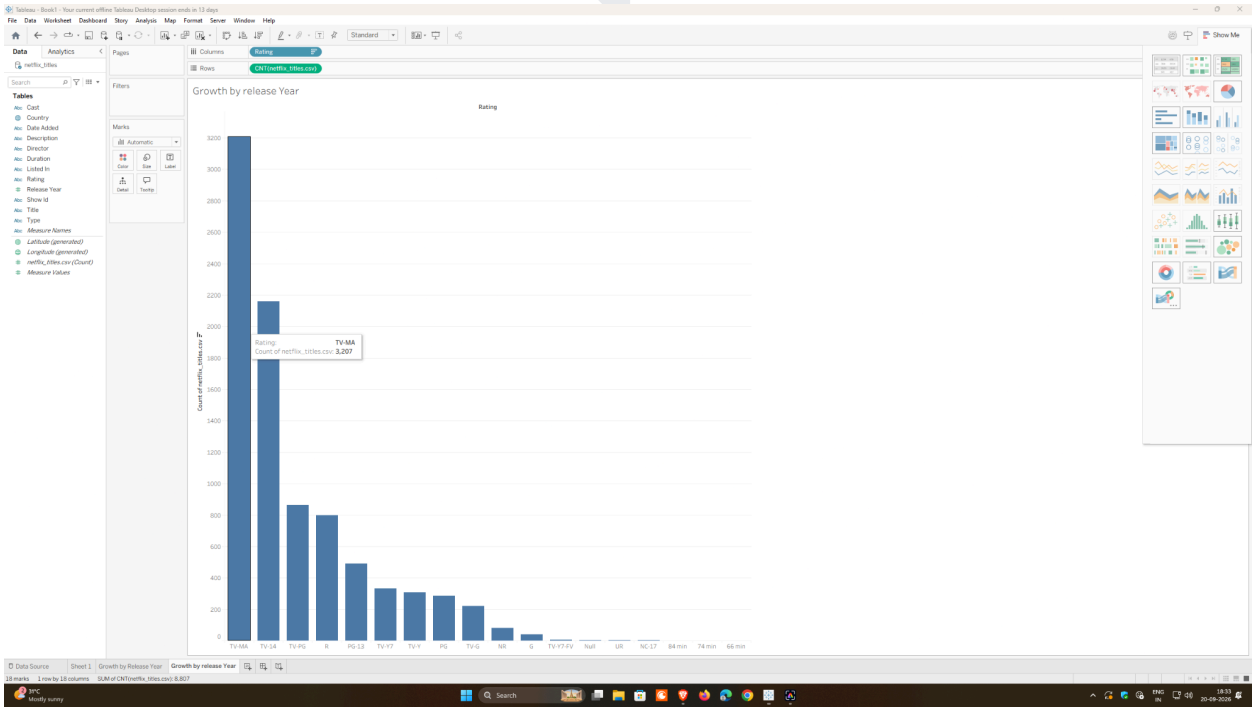
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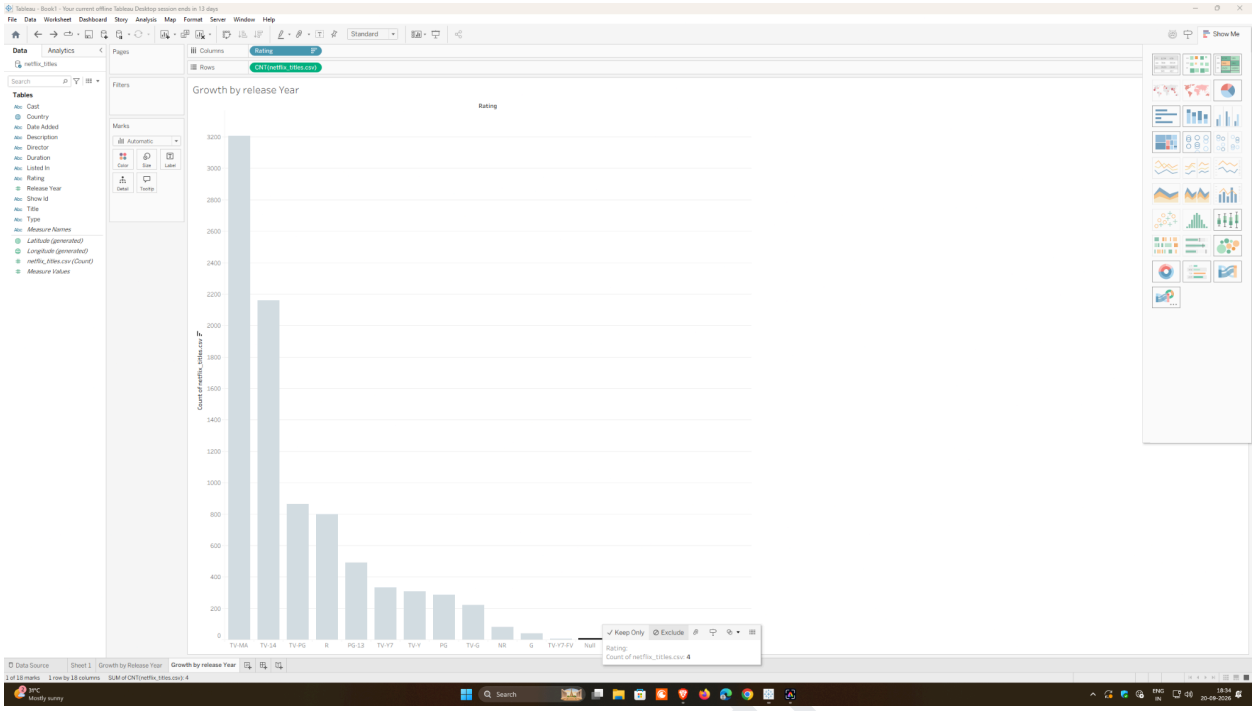
Sort the bars **descending by count** and filter out any **Null** ratings.



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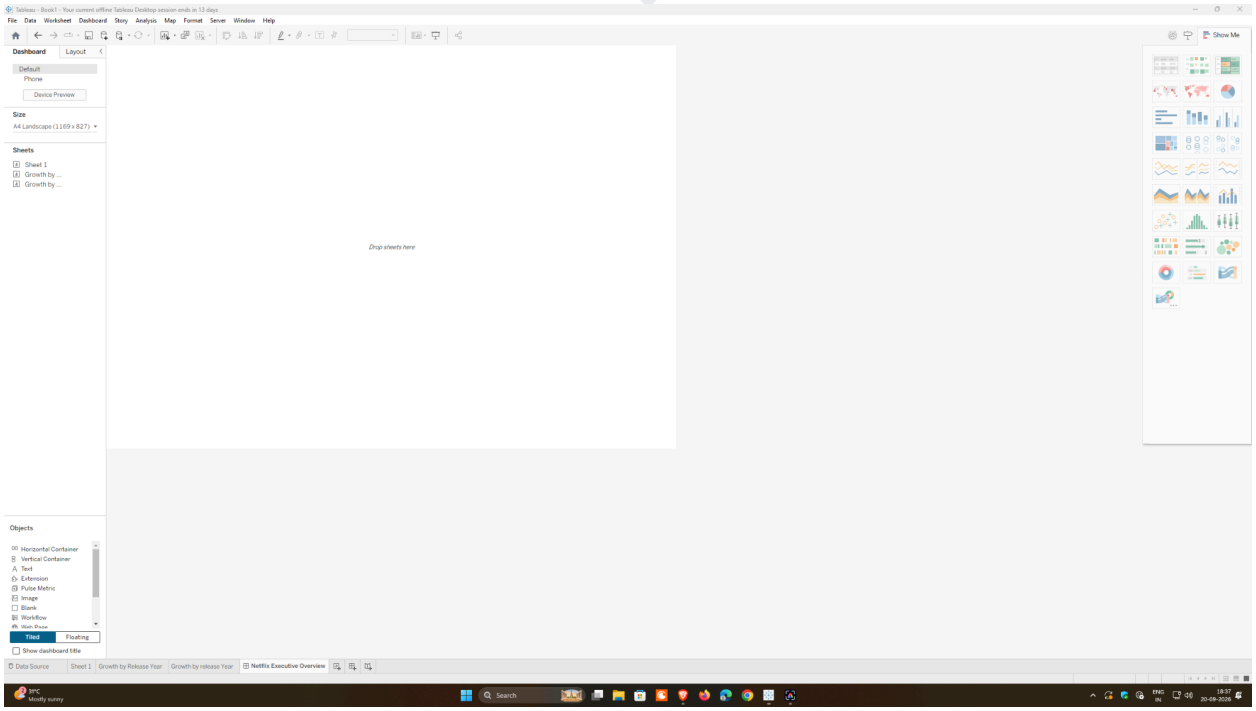
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Step 2: Assemble the Executive Dashboard

Create **Netflix Executive Overview** and set **Dashboard Size → Fixed → A4 Landscape (1169×827)** or **Letter Landscape (1100×850)**.

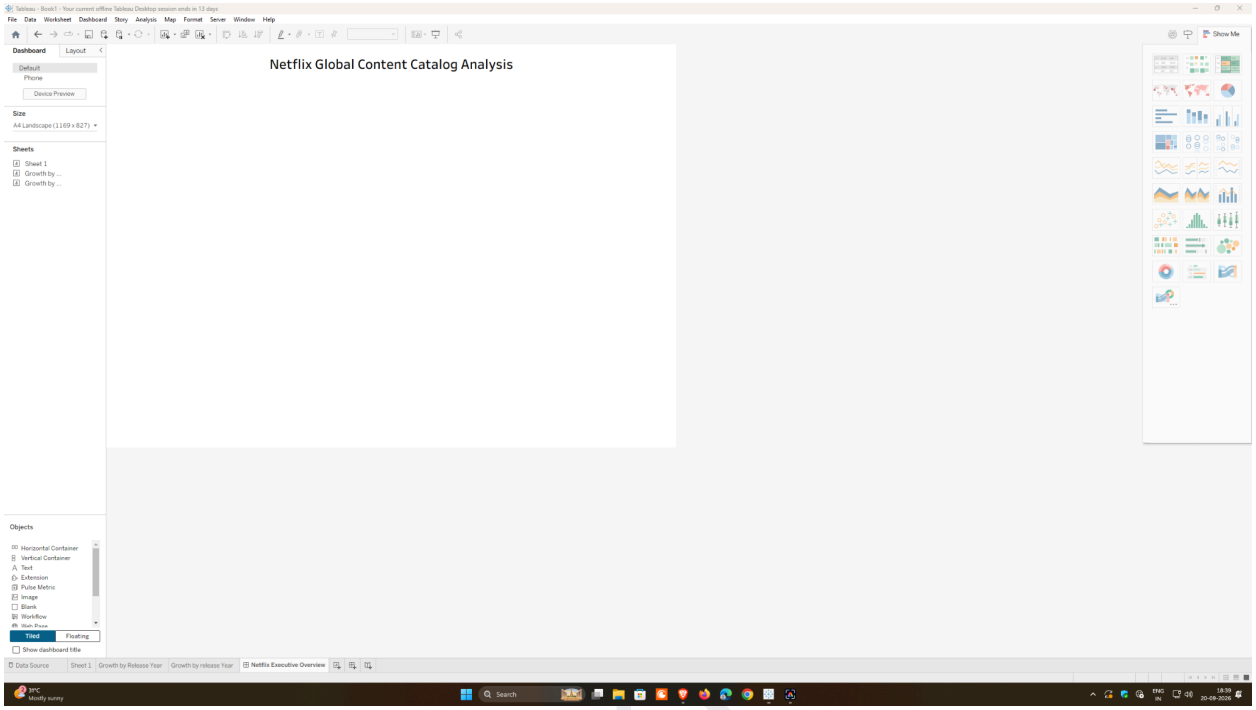


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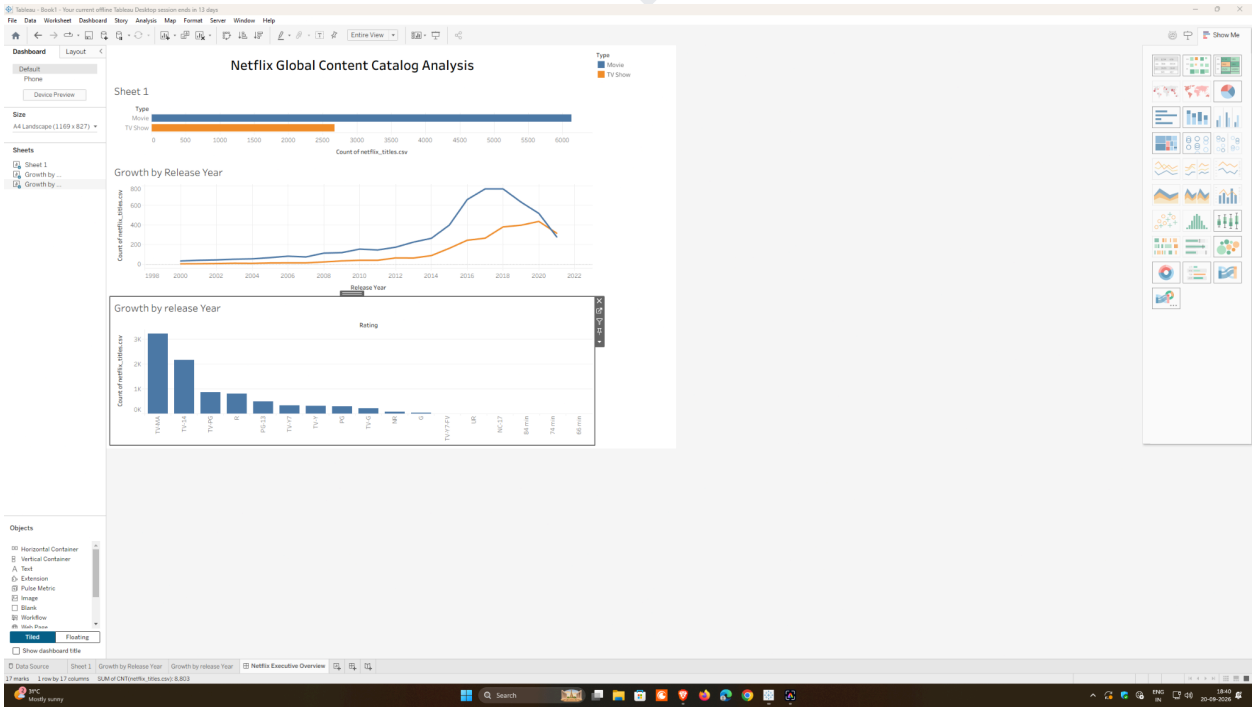
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Add a **Horizontal/Vertical Container**, then place **“Netflix Global Content Catalog Analysis”** at the top in **20 pt Bold**.



Place **Content Type Split** top-left, **Top 10 Ratings** top-right, and **Growth by Release Year** across the bottom.



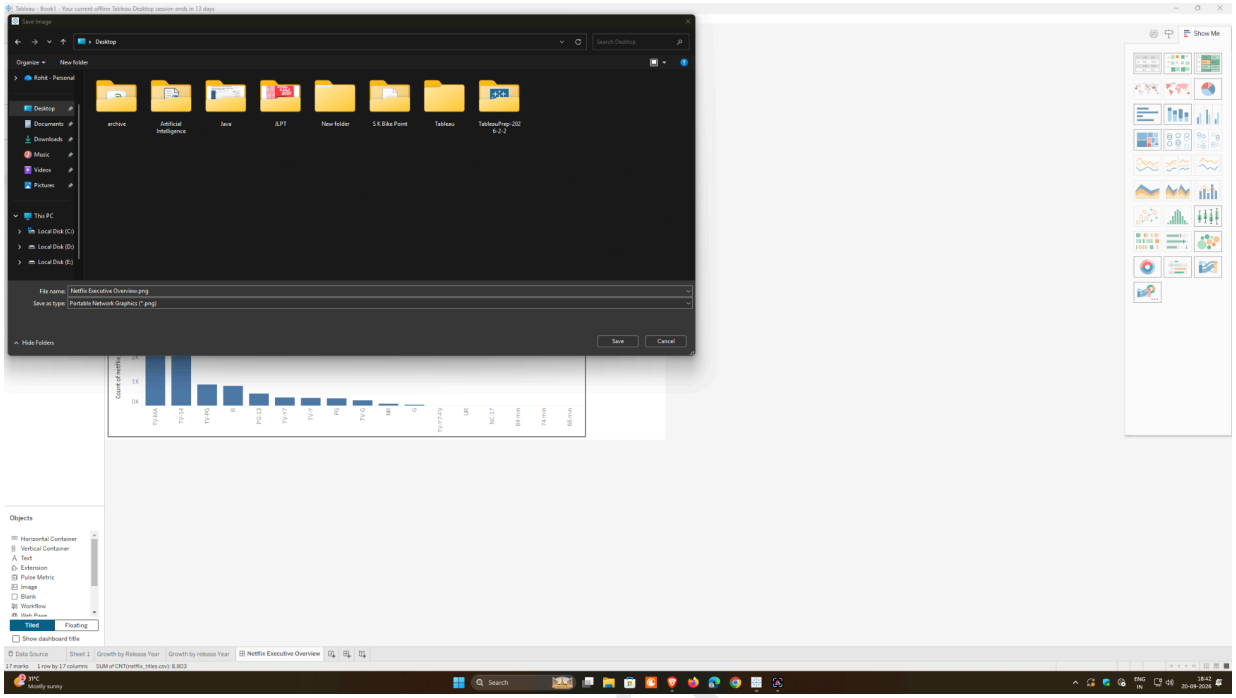
Step 3: Export as High-Resolution Image (Raster / Vector)

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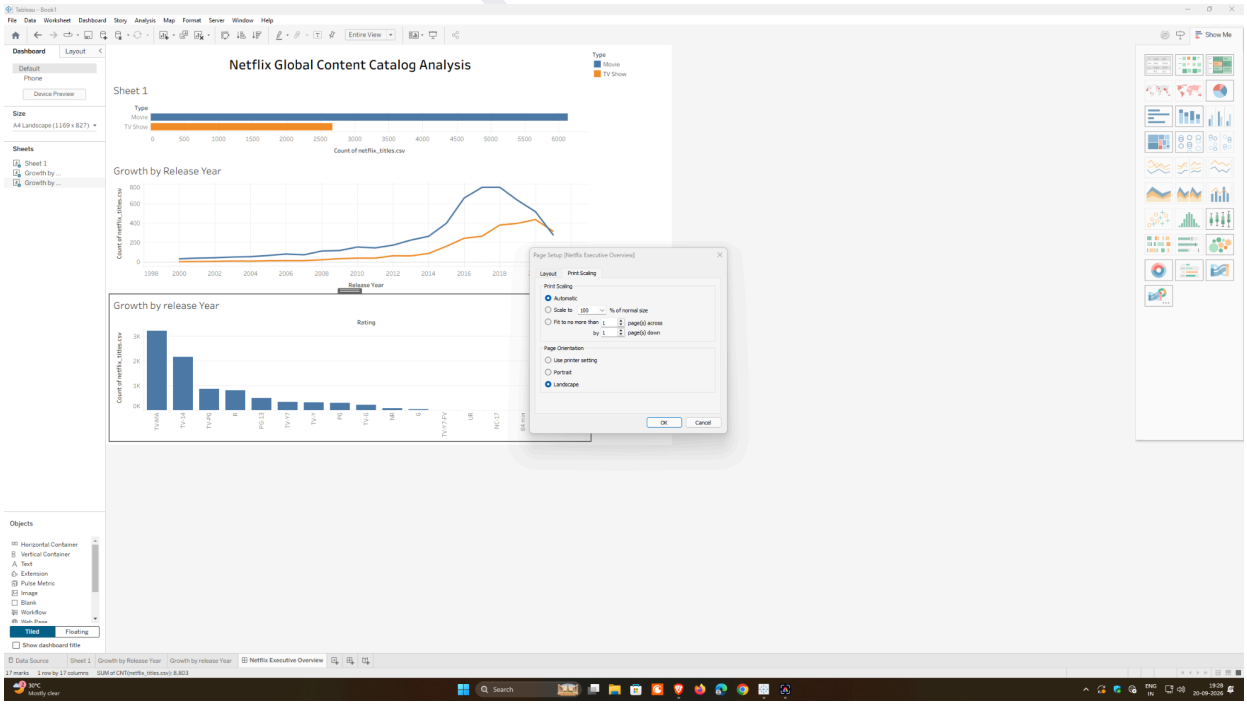
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Go to **Dashboard → Export Image**, name it **Netflix_Executive_Overview.png**, select **PNG Image (*.png)**, and click **Save**.



Step 4: Configure Print Setup and Export to PDF

Go to **File → Page Setup**, select **A4/Letter → Landscape → Fit to 1×1 page**, then click **OK**.

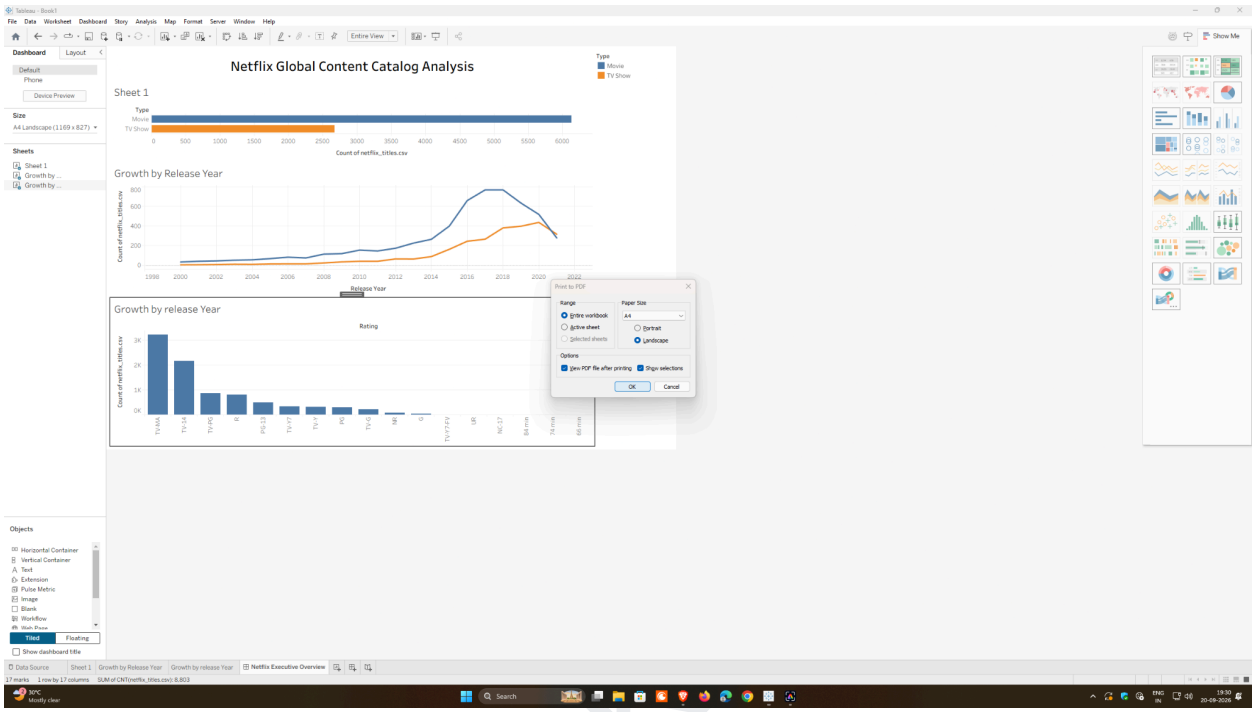


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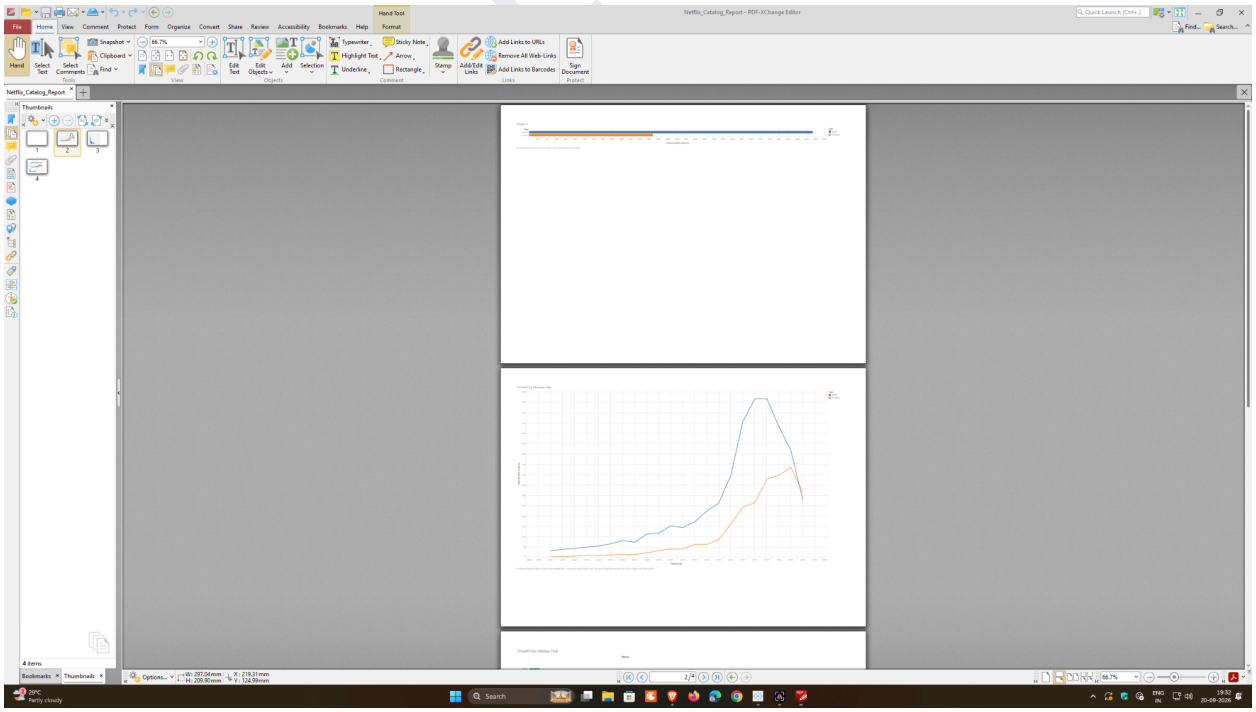
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Go to **File → Print to PDF → This Dashboard → A4/Unspecified → Landscape**, click **OK**, and save as **Netflix_Catalog_Report.pdf**.



Open **Netflix_Catalog_Report.pdf** and verify **clear output, no pixelation, and proper margin alignment**.



B: Use Presentation Mode in Tableau

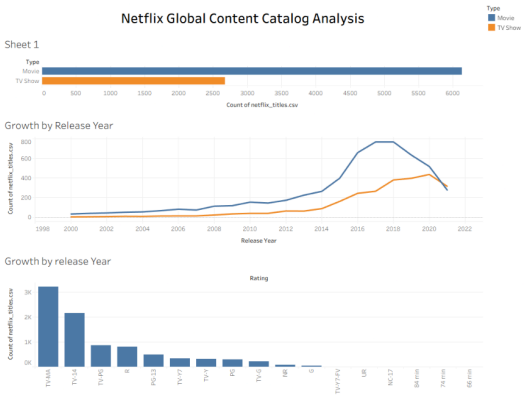
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Step 1: Launch Presentation Mode

1. Open your Netflix Executive Overview dashboard in Tableau Desktop.
2. Enter Presentation Mode using any of these methods:
   - Keyboard Shortcut: Press F7 (Windows) or Cmd + Enter (Mac).
   - Toolbar Icon: Click the Presentation Mode icon (a small projection screen icon) in the top application toolbar.
   - Menu Path: Go to the top menu and select Window --> Presentation Mode.



Step 2: Use On-Screen Presentation Controls

Once in full-screen mode, an unobtrusive toolbar appears at the bottom or top corner of the display:

Sheet Navigation: Use the left/right arrows on the toolbar (or the Page Up / Page Down keys) to cycle through all worksheets, dashboards, and stories in your workbook.

Live Interactive Filtering: Click on any bar or line mark (e.g., clicking Movies in the split chart) to demonstrate real-time cross-filtering.

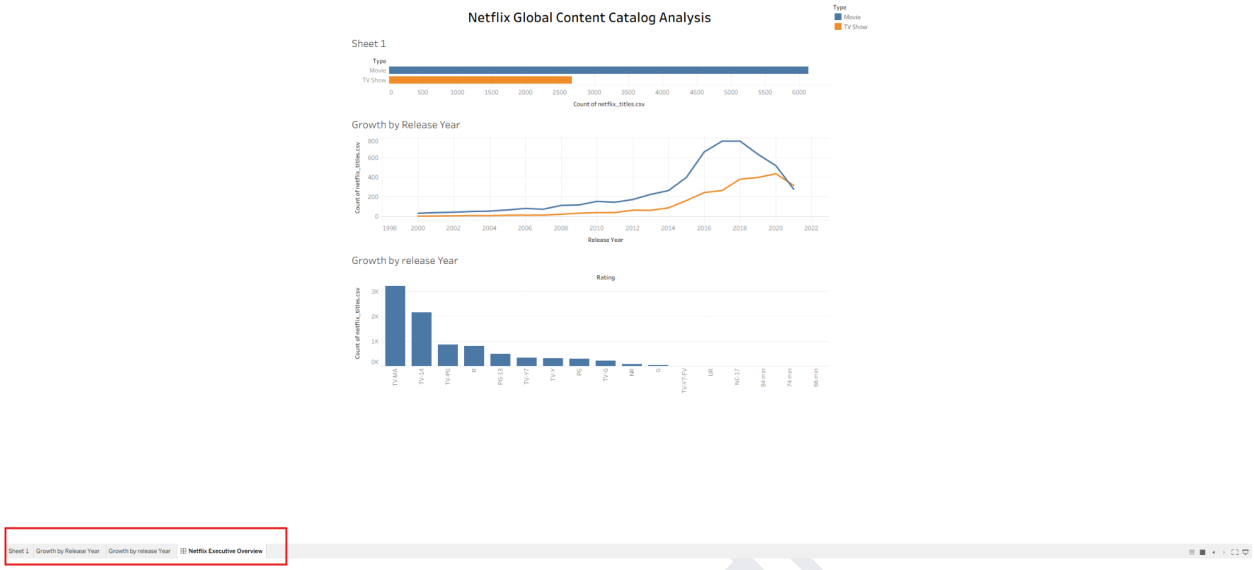
Interactive Tooltips: Hover over data points to present granular numbers on demand without cluttering the main slide.

Reset / Revert: Click the Revert arrow on the presentation toolbar to reset all filters back to their initial state.

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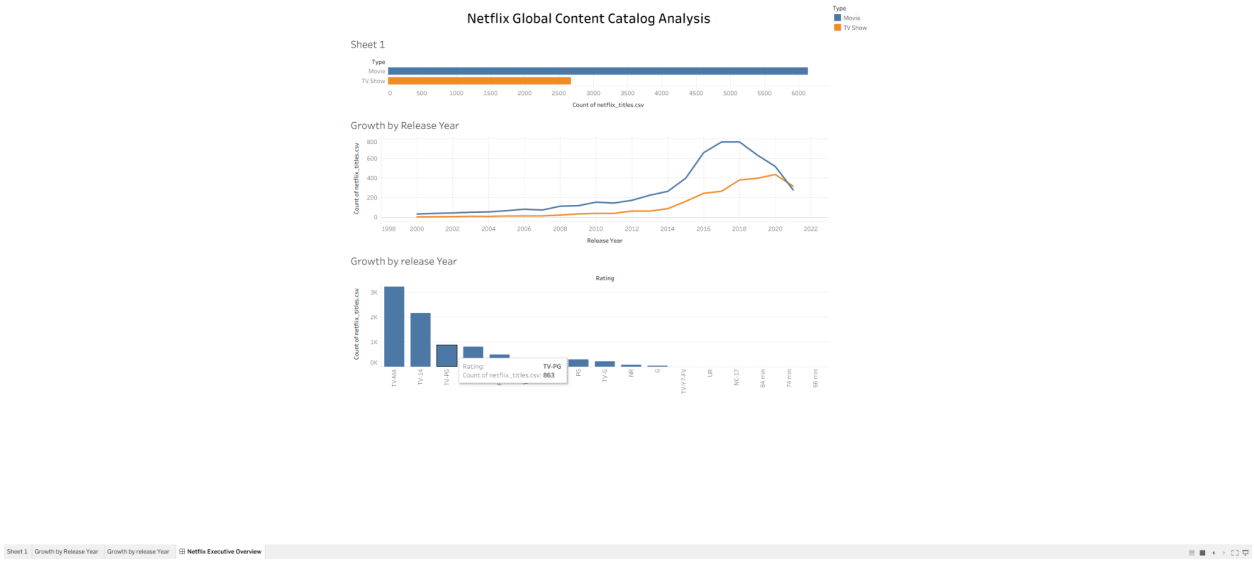
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Step 2: Live Navigation and Interactivity During Presentation
While in Presentation Mode, test the following live controls:

1. Hover Exploration:

- Move your mouse over individual data marks (e.g., specific bars in Content Type Split or peak data points in Growth by Release Year).
- Confirm that tooltips render cleanly with exact counts and dates.



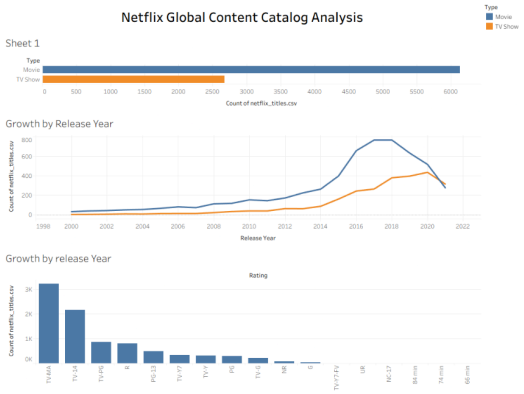
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2. Interactive Filtering:

- If quick filters or legends are placed on the dashboard, click to filter or
- highlight categories live.

3. Multi-Sheet Navigation:

- Move your mouse to the bottom-left corner to reveal the subtle navigation toolbar.
- Use the Next Sheet ( ) and Previous Sheet () arrows (or the Page Up / Page Down keys) to cycle between individual worksheets and dashboards seamlessly.





Step 3: Exit Presentation Mode
Press the Esc key on your keyboard (or click the Exit Presentation Mode icon in the bottom-left toolbar) to return to standard authoring mode.

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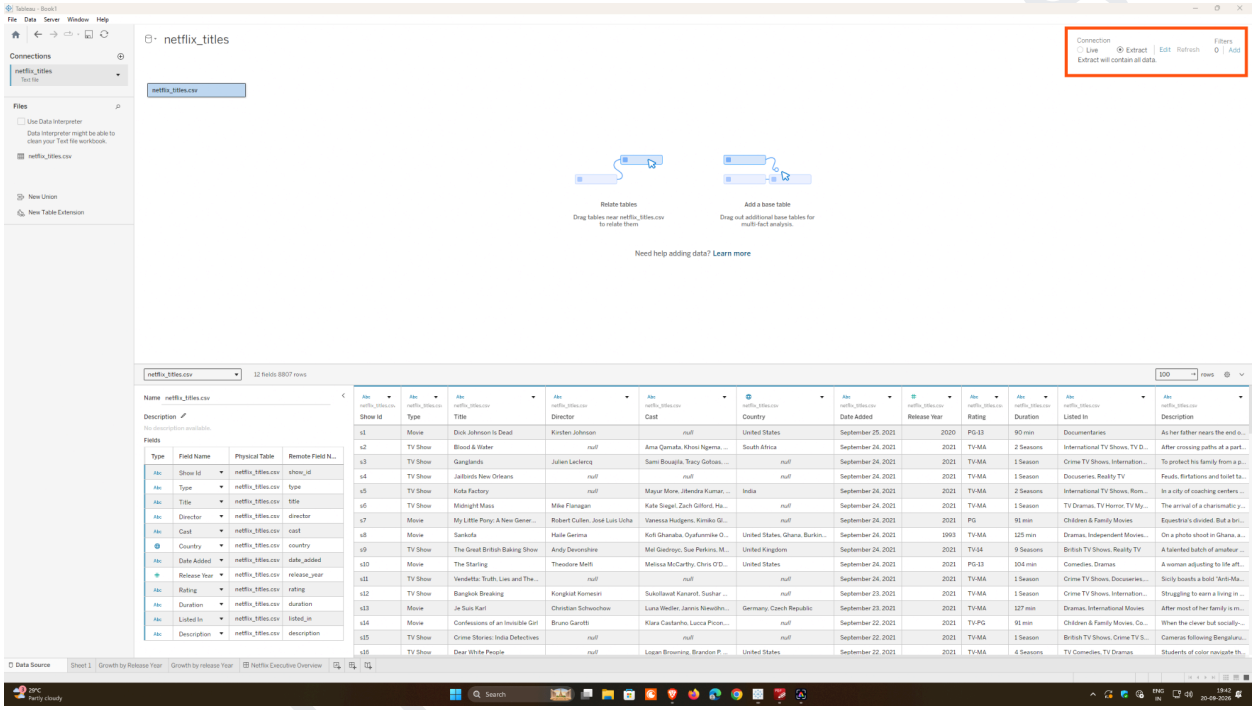
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C: Publish Dashboards to Tableau Public

Step 1: Confirm Data Extract Status

1. Check the top-left Data Pane where netflix_titles is listed:
   - If you see two linked cylinders (🛢️🛢️), the data is already extracted. Proceed to Step 2.
2. If you see a single cylinder (🛢️):
   - Go to the Data Source tab in the bottom-left corner.
   - In the top-right corner, select the Extract radio button.



   - Click back onto your Netflix Executive Overview dashboard tab

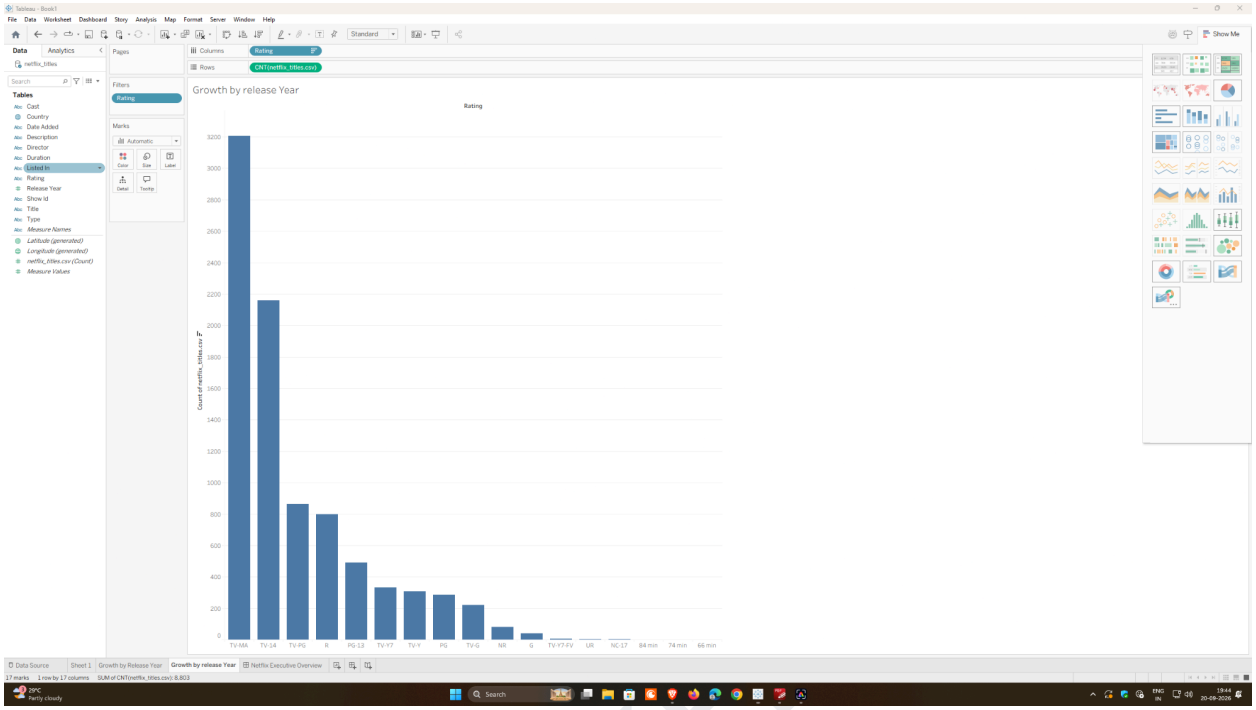
Step 2: Prune Unused Columns to Minimize File Footprint

1. In the Data Pane, click the drop-down menu icon (small down-arrow next to the search bar).
2. Select Hide All Unused Fields.
   - Outcome: Columns not used in your visualizations or calculations (e.g., full text descriptions or raw cast lists, if unused) will be excluded from the package, reducing disk space and load times.

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Step 3: Pin the Default Landing Tab

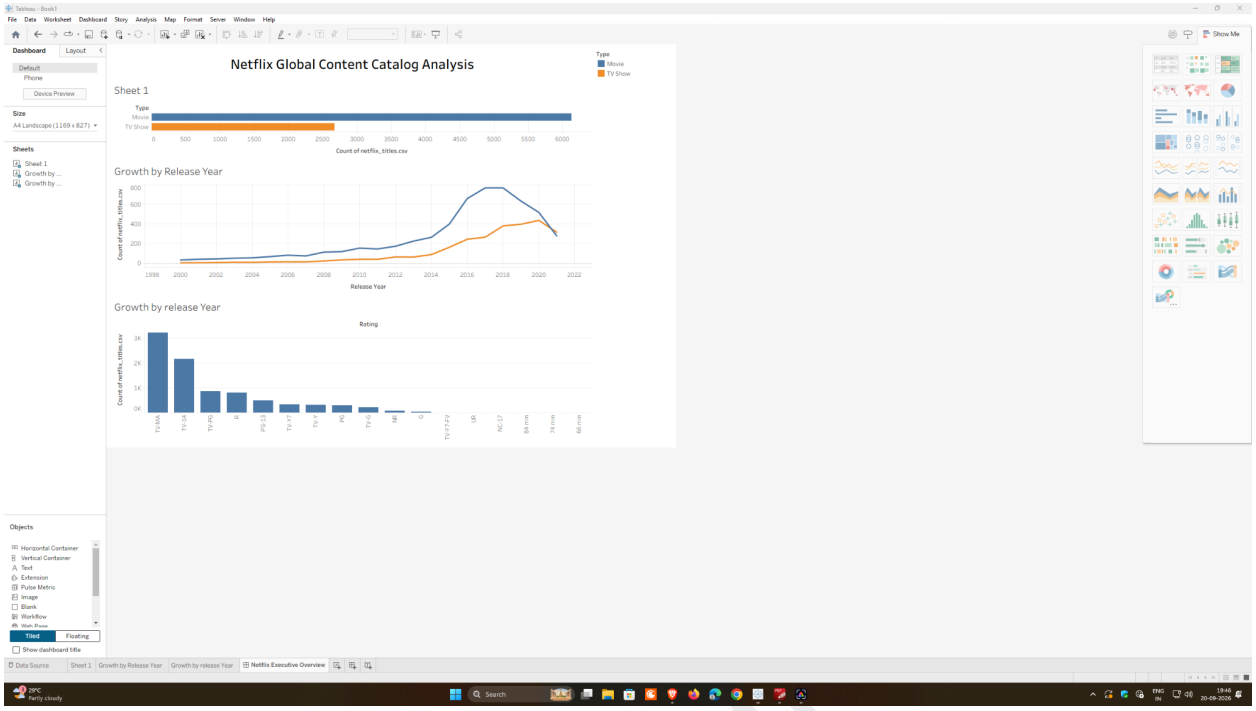
1. Click on the Netflix Executive Overview tab at the bottom so it is active.
2. In the left-hand Dashboard panel, ensure Size is set to Fixed Size (e.g., 1366 x 768 px or A4 Landscape).
    - Why: Fixed sizing prevents web browsers on different screen resolutions from distorting the chart layout or causing unwanted scrollbars.

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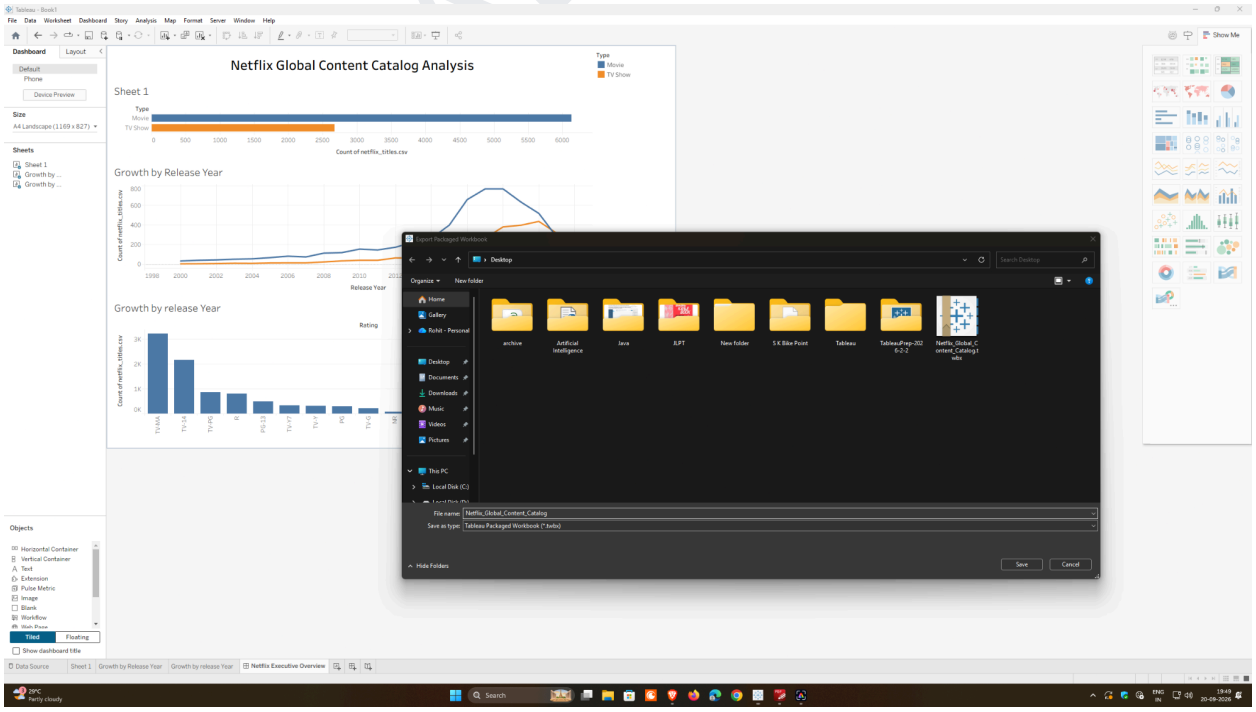
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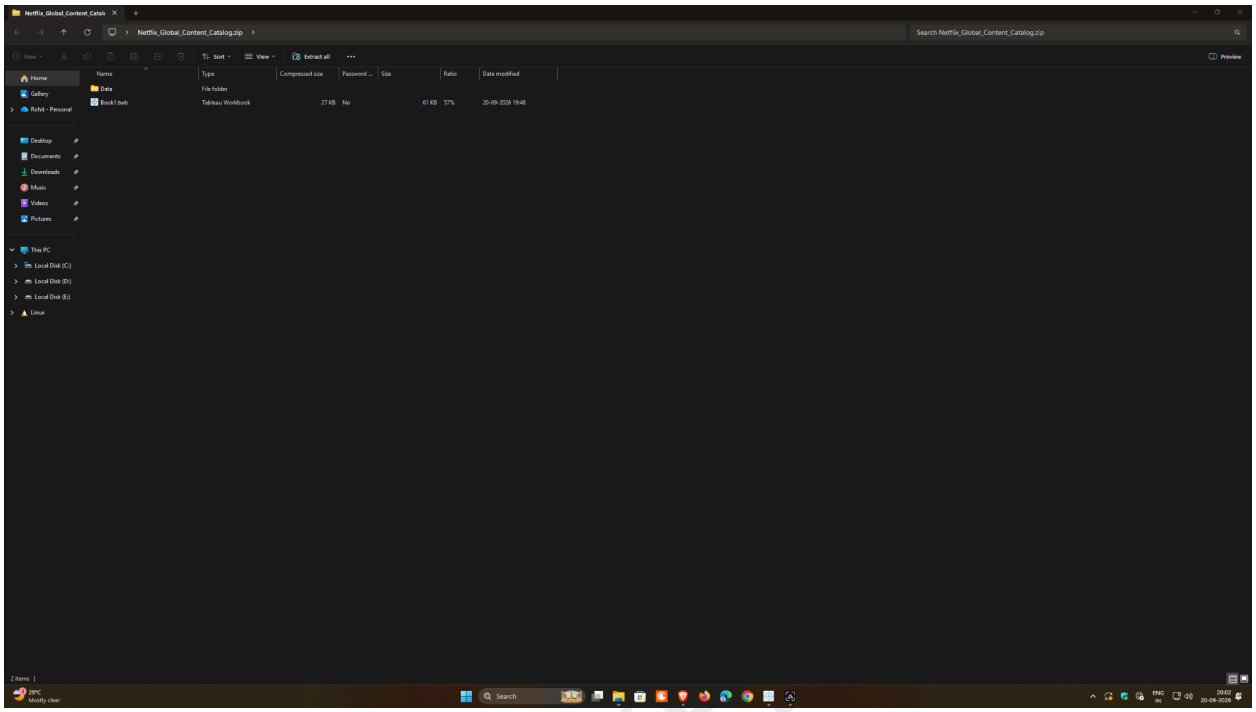
Step 4: Export the Standalone Packaged Workbook (.twbx)

1. Go to the top menu bar and click File --> select Export Packaged Workbook... (or File --> Save As...).
2. In the file save window: o File name: Netflix_Global_Content_Catalog.twbx
o Save as type: Select Tableau Packaged Workbook (*.twbx)
3. Click Save.



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Step 5: Verify Archive Integrity Locally



4. This confirms the workbook can be moved to any computer or shared via USB/email without needing netflix_titles.csv.

D: Share Dashboards via Tableau Server / Tableau Cloud

Step 1: Clean Up Workbook Tabs (Sheet Governance)

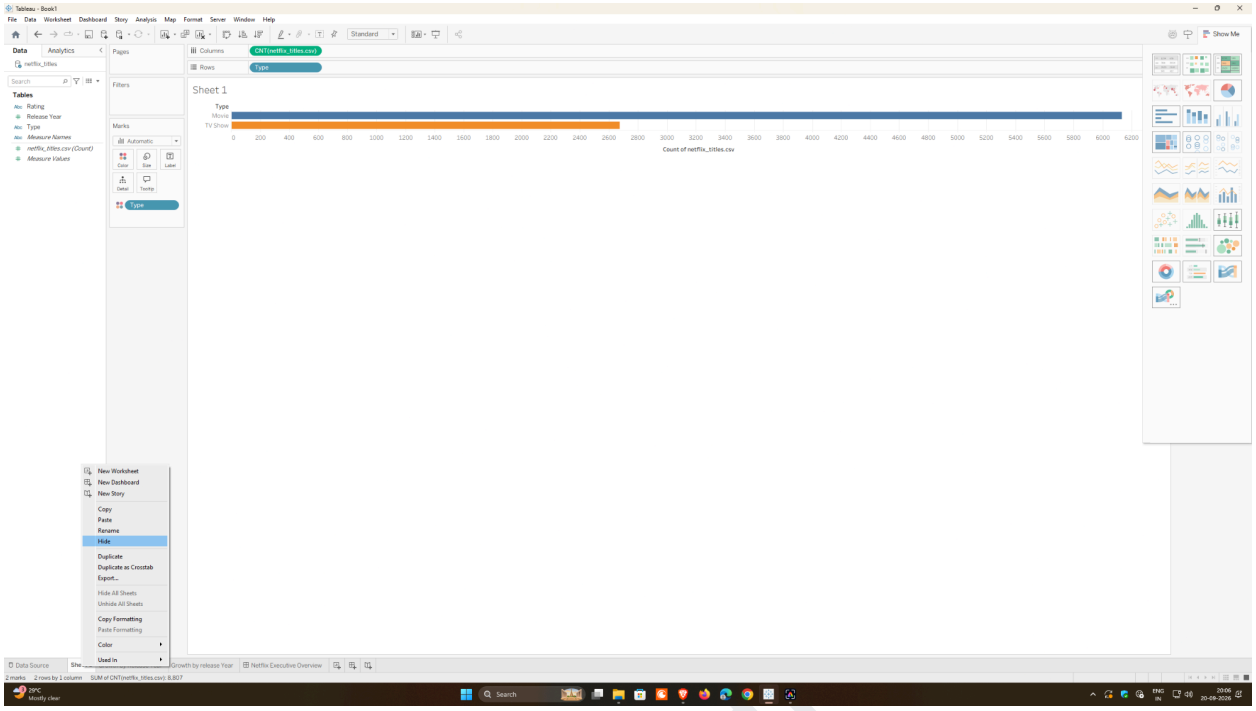
1.Hide each underlying worksheet:

- Right-click the Content Type Split tab --> select Hide Sheet.
- Right-click Growth by Release Year --> select Hide Sheet.
- Right-click Top 10 Ratings --> select Hide Sheet.
- (Note: If "Hide Sheet" is grayed out, ensure the sheet is currently placed inside your Netflix Executive Overview dashboard)

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Verify the Layout: Look at the bottom tab bar. Only the Netflix Executive Overview dashboard tab should now be visible.

Step 2: Configure Navigation and Sheet Tabs Settings

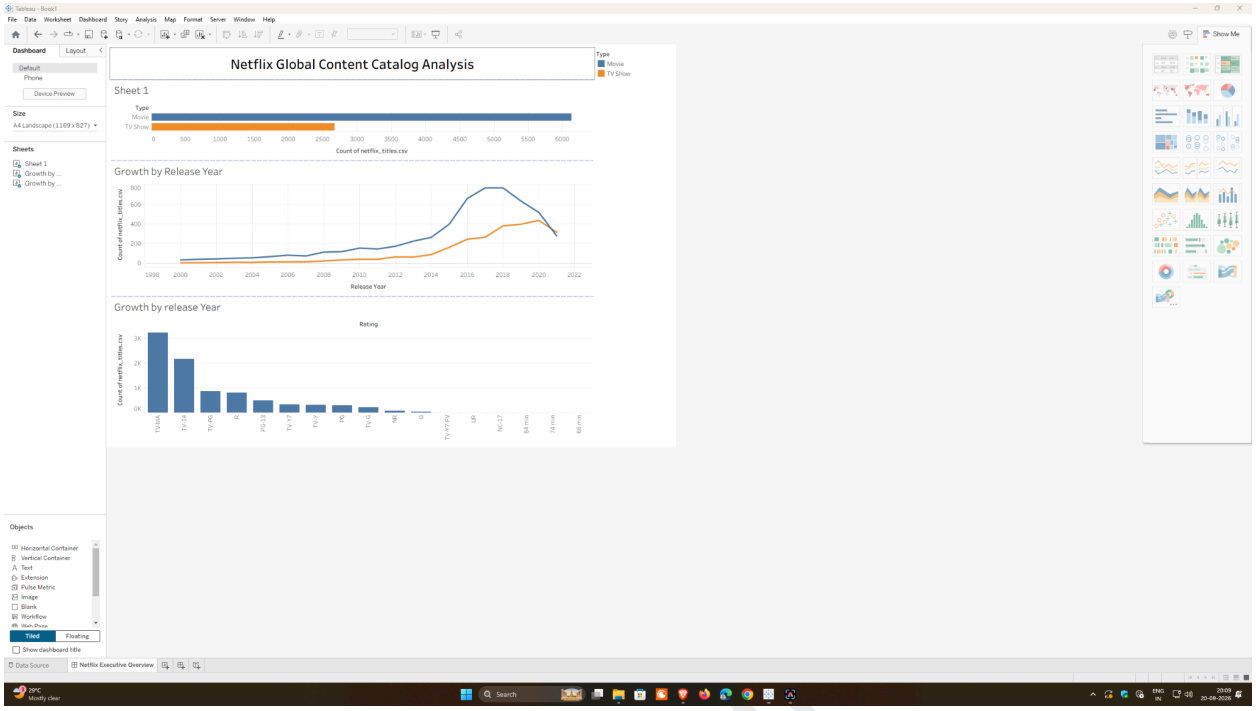
1. Go to the top application menu bar and click Window.
2. Look at the Show Sheet Tabs toggle:
    - If you have only one executive dashboard, you can leave it unchecked or keep it clean so the dashboard spans the entire interface.
3. Verify that the window controls are clean and unobstructed by unused toolbars.

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Step 3: Understand the Tableau Reader Distribution Mechanism:
When Tableau Server or Tableau Cloud is unavailable, organizations distribute dashboards via Tableau Reader:

What is Tableau Reader?

- A free, standalone desktop viewer application for Windows and macOS.
- Downloadable at no cost by any stakeholder or client without a license key.

Consumer Capabilities in Tableau Reader:

- **Interactivity**: Viewers can click marks, filter categories, hover for tooltips, and change parameters.
- **Zero Accidental Edits**: Viewers cannot move charts, alter formulas, break calculations, or modify visual aesthetics.
- **File Requirement**: Tableau Reader opens only .twbx (packaged) workbooks with extracted data, ensuring stakeholders never experience "data source not found" errors.

Step 4: Simulate Consumption via Tableau Reader / Fresh Session
To verify that your workbook functions as a true enterprise-ready deliverable:
1. In Tableau Desktop, press Ctrl + S (or go to File $\rightarrow$ Save) to save your packaged workbook (Netflix_Global_Content_Catalog.twbx).
2. Close your current Tableau Desktop window.
3. Double-click the saved Netflix_Global_Content_Catalog.twbx file in your local file explorer to reopen it.
4. Audit the File:

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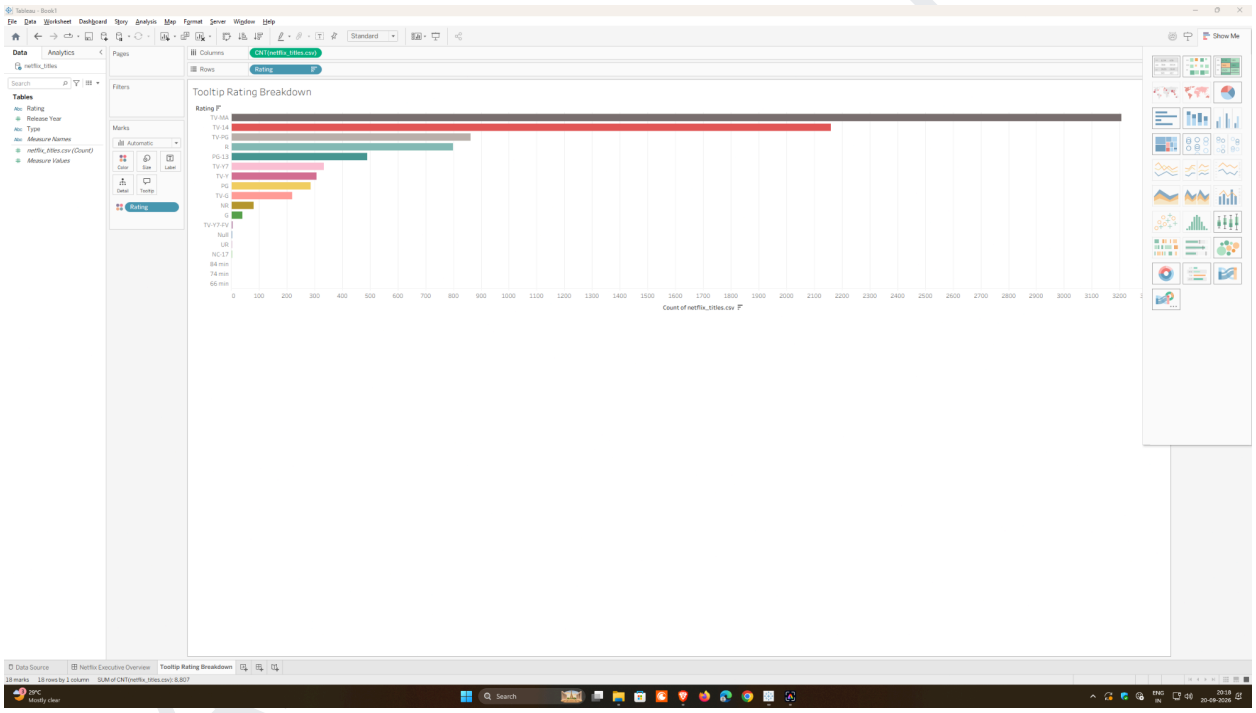
- Does it open directly to the Netflix Executive Overview?
- Are the underlying calculation/raw sheets hidden from view?
- Does it open instantly without asking for database passwords or CSV file paths?

5. If yes to all, your deliverable is structured correctly for offline distribution.

E: Enable User Interaction on Shared Dashboards

Step 1: Create the Detail Sheet for Viz in Tooltip

1. Create a new worksheet named: Tooltip Rating Breakdown.
2. Drag Rating onto the Rows shelf.
3. Drag netflix_titles.csv (Count) onto the Columns shelf.
4. Drag Rating onto the Color tile on the Marks card.
5. Sort the bars in descending order.



Step 2: Embed the Viz into the Main Tooltip

1. Return to your Content Type Split worksheet.

2. On the Marks card, click the Tooltip tile.

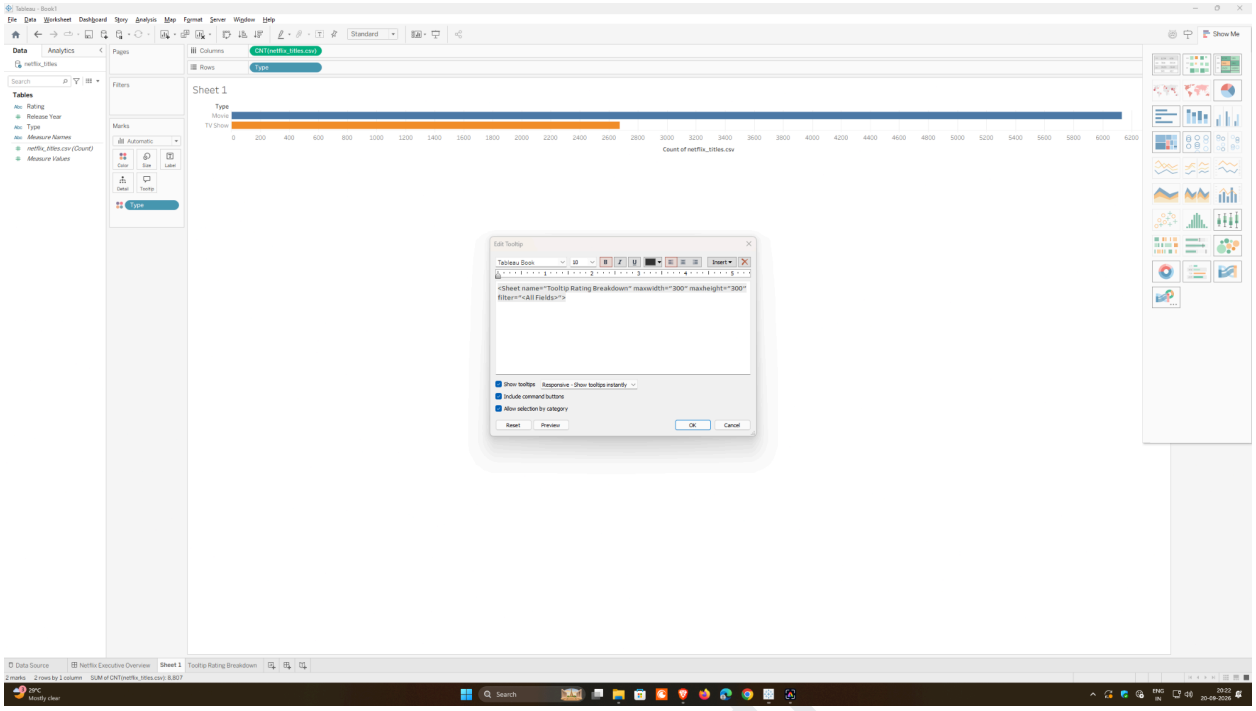
3. In the Edit Tooltip dialog: o Delete default text if desired, then click Insert in the top-right corner of theeditor.

- Select Sheets --> Tooltip Rating Breakdown. o Tableau inserts the code tag: <Sheet name="Tooltip Rating Breakdown" maxwidth="300" maxheight="300" filter="<All Fields>">

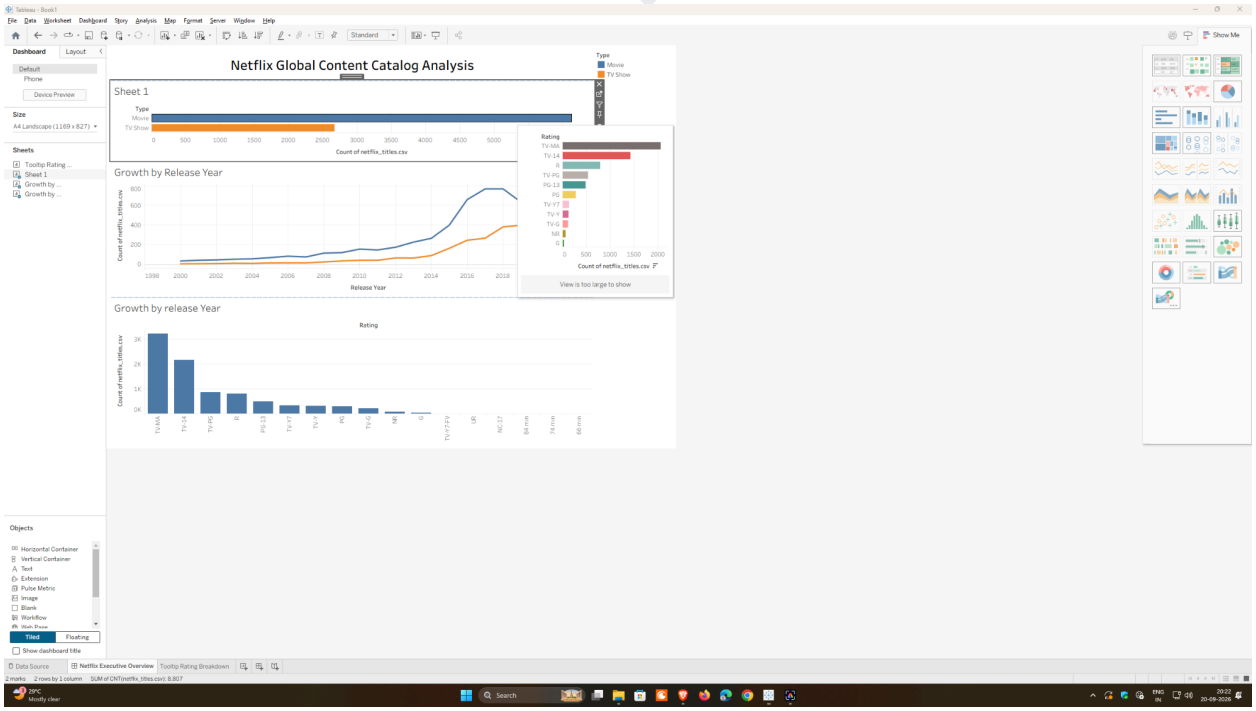
4. Click OK.

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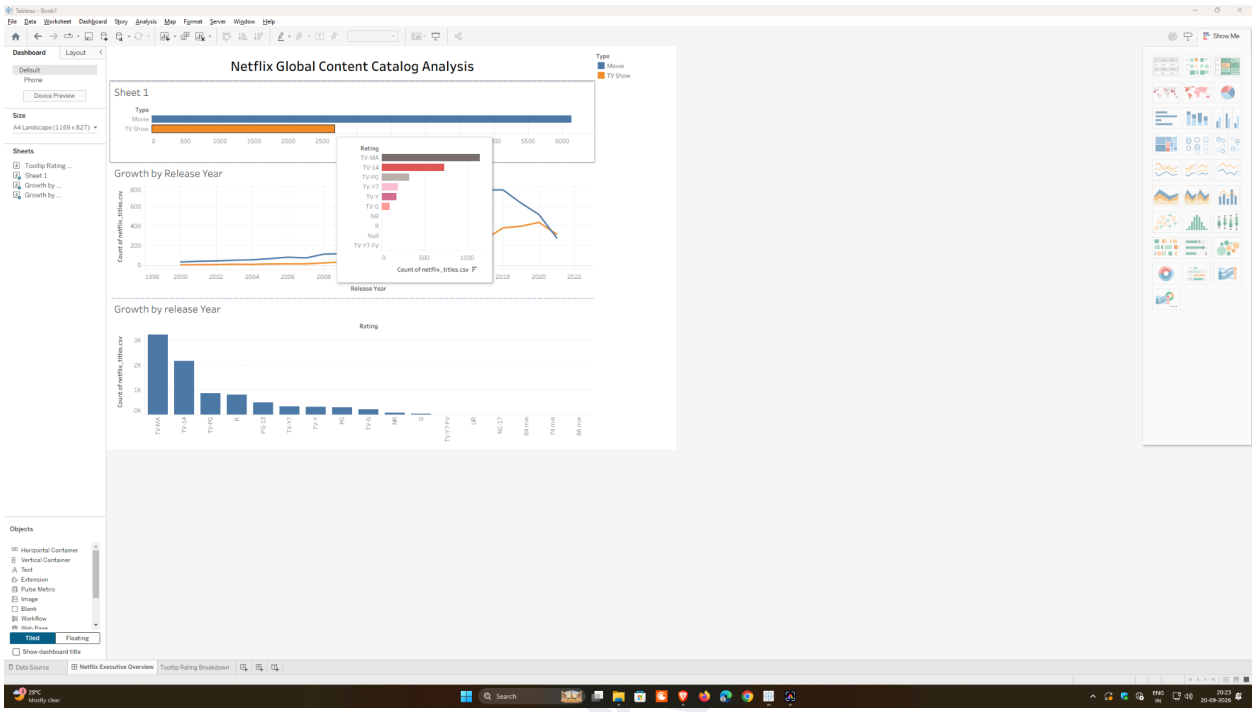


5. Switch back to your Netflix Executive Overview dashboard and test it:
    - Hover your mouse over the Movie bar: you will see a mini-bar chart inside the tooltip showing the rating distribution for movies only.



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- Hover over the TV Show bar: the mini-bar chart instantly changes to show TV show ratings (TV-MA, TV-14, etc.).



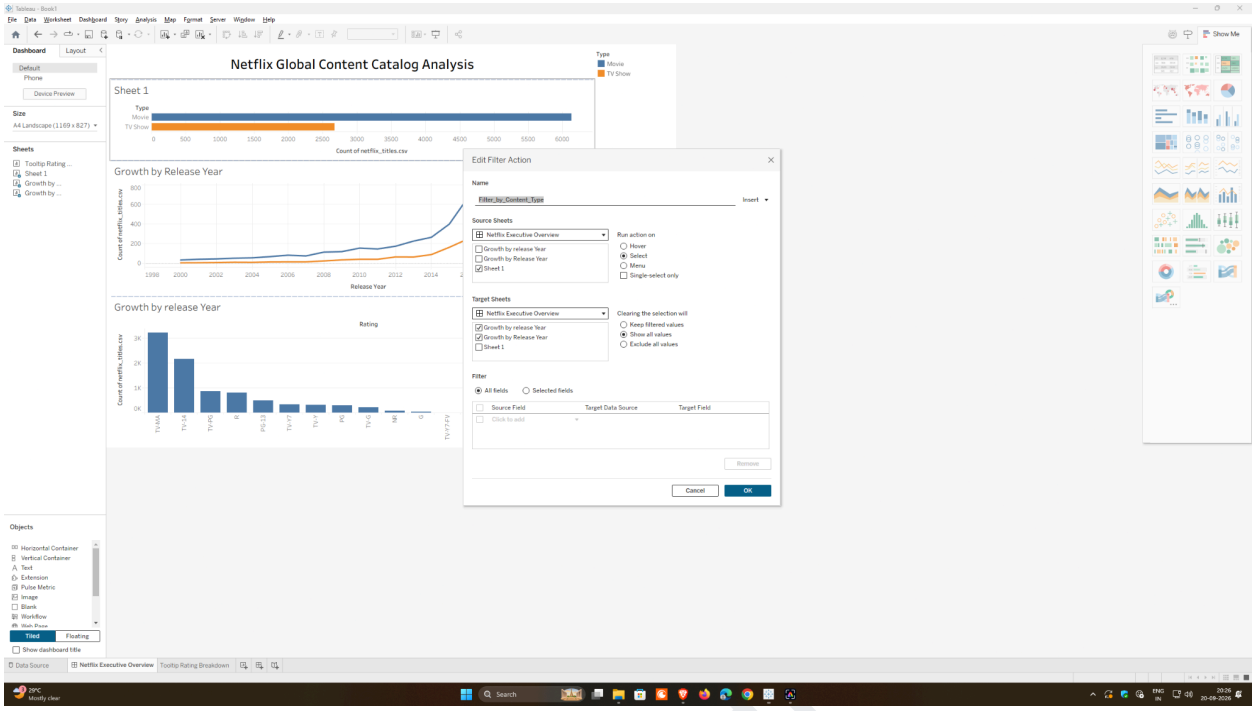
Step 3: Configure Dashboard Filter Actions (Cross-Filtering)

1. Open your Netflix Executive Overview dashboard.
2. From the top menu bar, select Dashboard --> Actions... (or press Ctrl + Shift + A).
3. In the Actions dialog, click Add Action --> select Filter....
4. Configure the Filter Action:
   - Name: Filter_by_Content_Type
   - Source Sheets: Check Content Type Split. (Sheet 1)
   - Run action on: Select Select (triggers on click).
   - Target Sheets: Check Growth by Release Year and Top 10 Ratings.
   - Clearing the selection will: Select Show all values.
   - Target Filters: Leave as All Fields.
   - NOTE: Name of Sheets can be different.

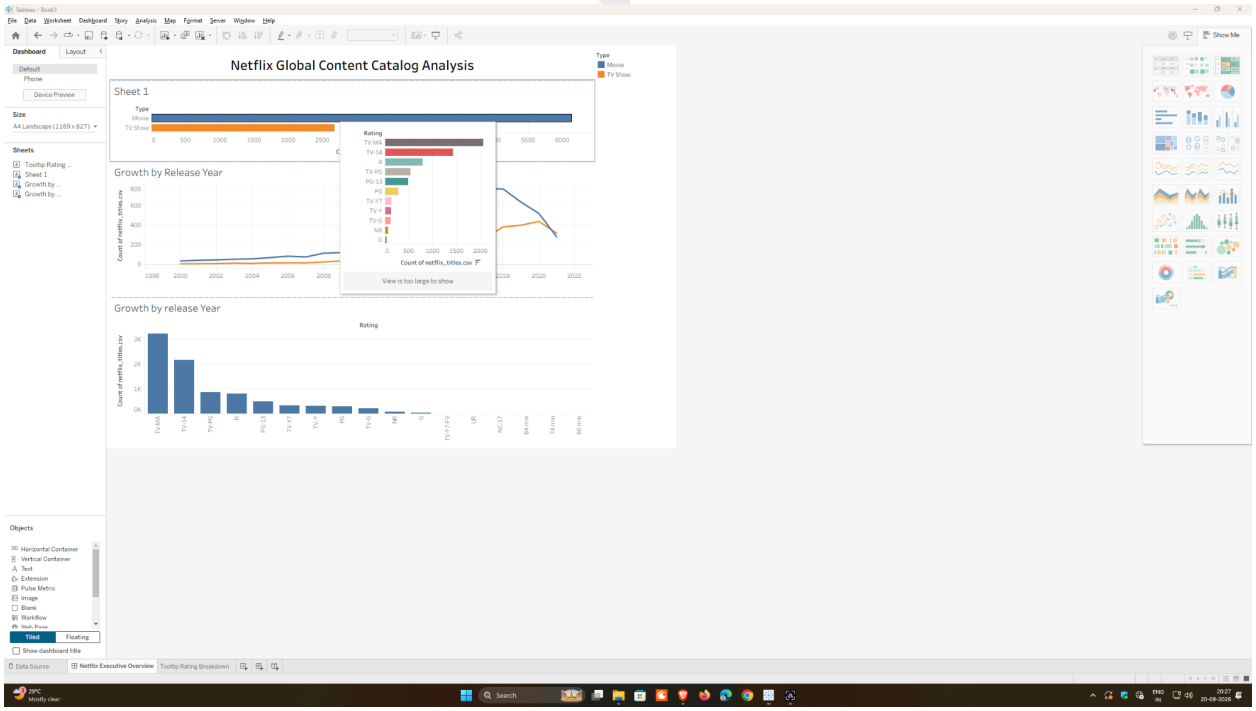
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5. Click OK, then click OK again.
6. Test the Filter: Click the Movie bar; the line chart instantly updates to show release year trends exclusively for movies. Click it again to clear the filter.



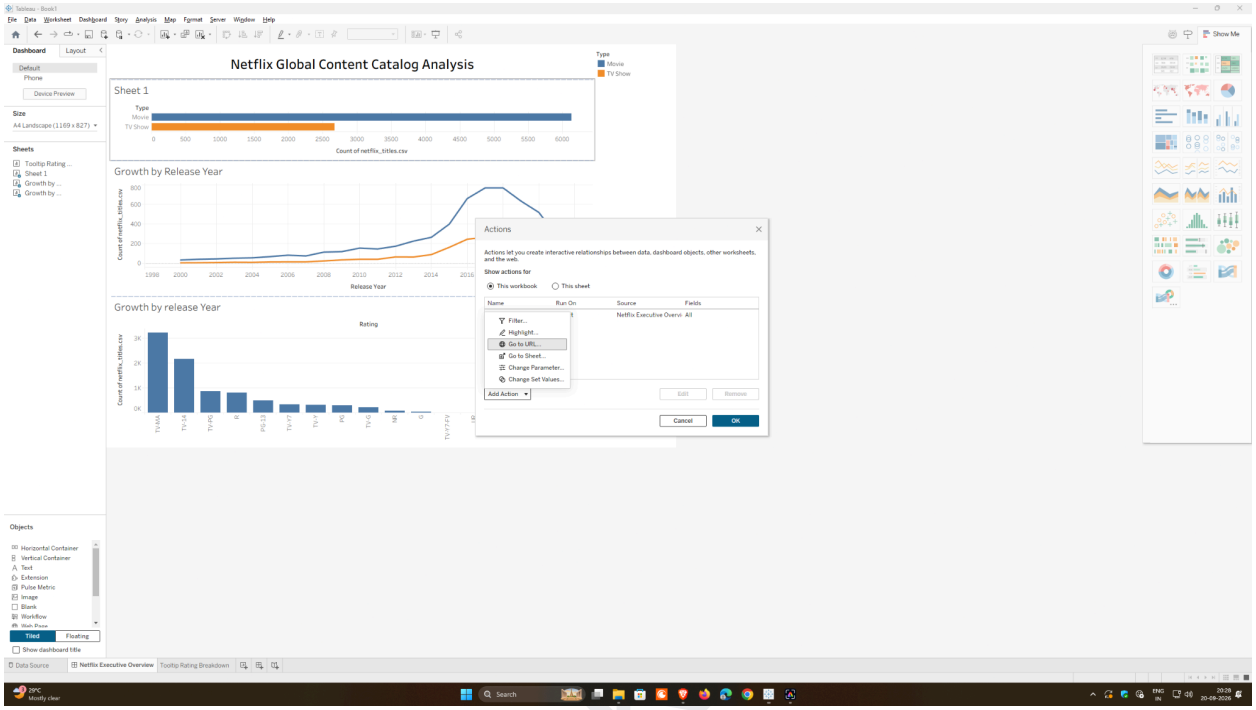
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Step 4: Add a Contextual Web URL Action

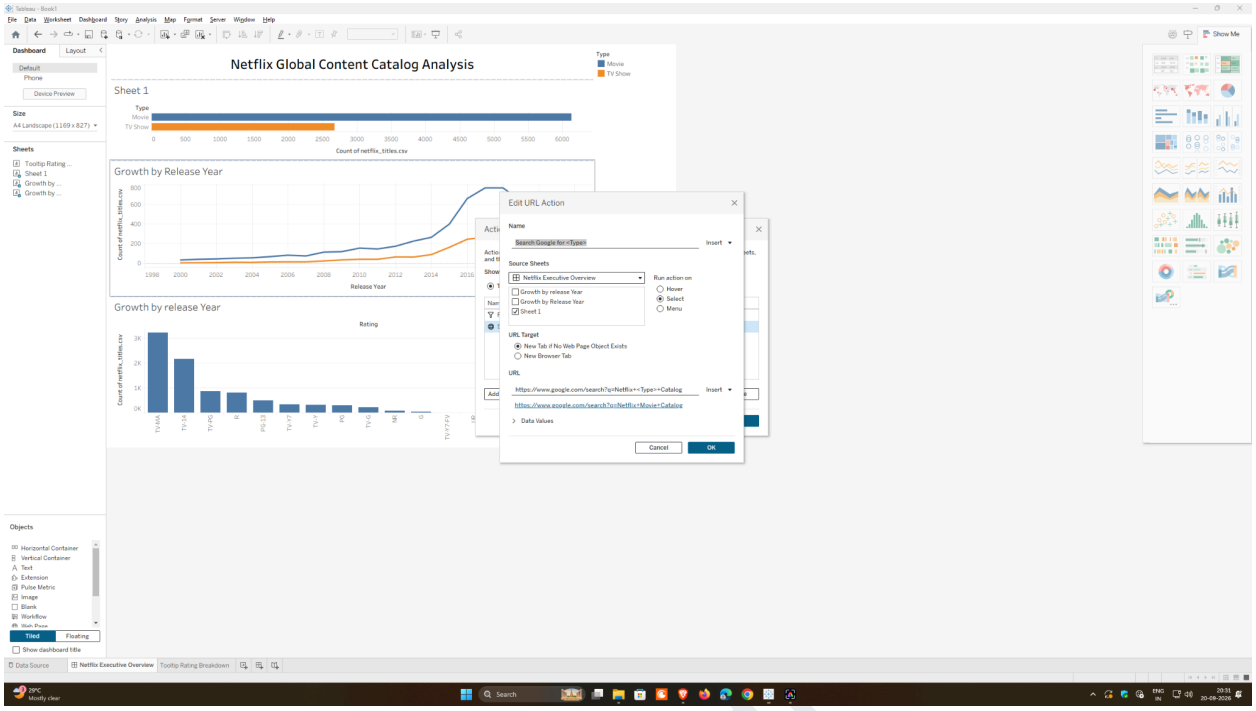
1. Go to Dashboard --> Actions... --> click Add Action --> select Go to URL....



2. Configure the URL Action: o Name: Search Google for <Type>
   - Source Sheets: Check Content Type Split.
   - Run action on: Select Menu (adds a clickable link inside the tooltip).
   - URL: Enter: https://www.google.com/search?q=Netflix+<Type>+Catalog o (Click the small arrow next to URL to insert the <Type> dimension dynamically).
3. Click OK, then click OK to close.

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4. Hover over any bar and click the hyperlink menu item to verify that the browser opens the dynamic search query.

F: Manage Permissions and Access Control

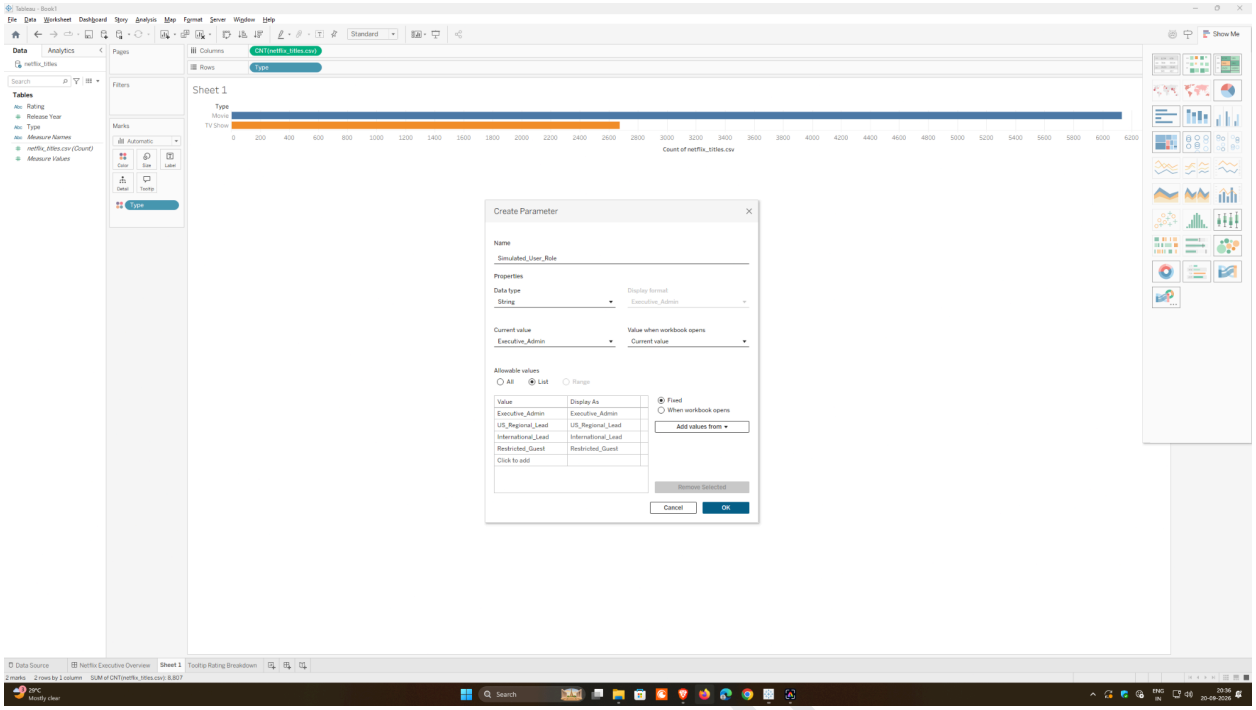
Step 1: Create the User Persona Simulation Parameter

1. In the Data Pane on the left, click the small dropdown arrow next to the search bar (or right-click an empty space) --> select Create Parameter....
2. Configure the parameter properties:
   - Name: Simulated_User_Role
   - Data type: String
   - Allowable values: Choose List
   - Add four simulated corporate identities:
     1. Executive_Admin
     2. US_Regional_Lead
     3. International_Lead
     4. Restricted_Guest
   - Current value: Set to Executive_Admin

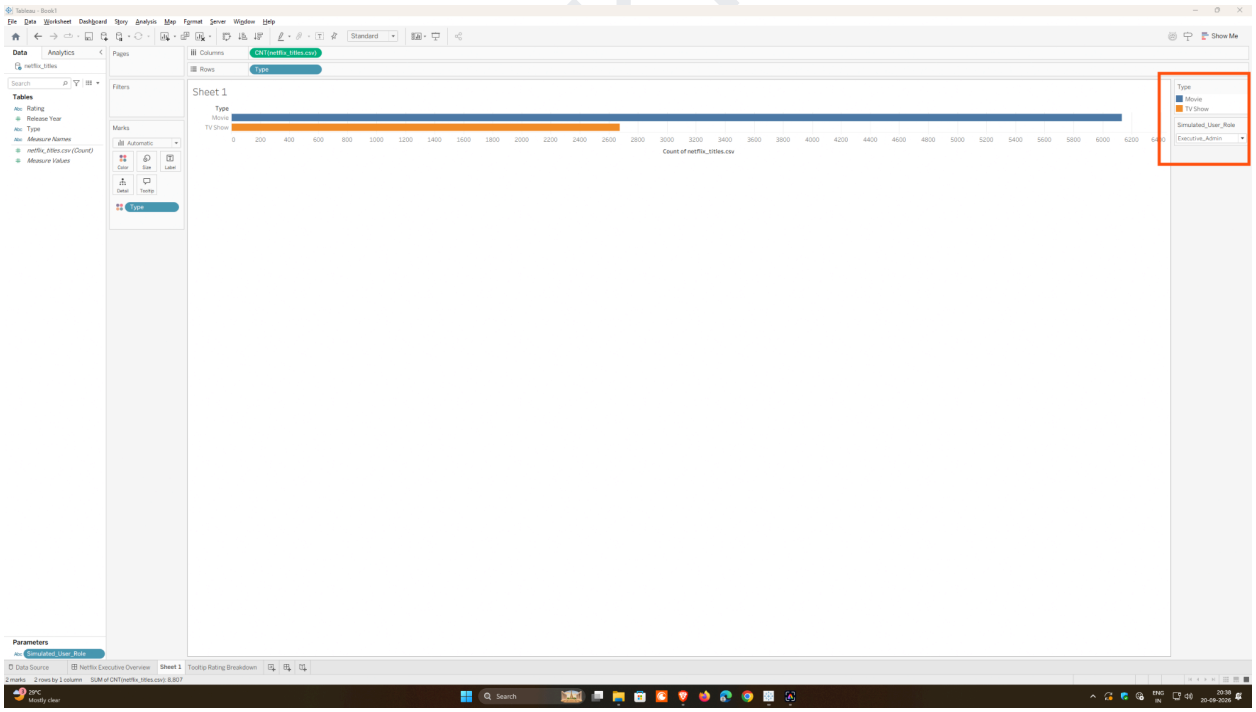
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4. In the Data Pane under Parameters, right-click Simulated_User_Role --> select Show Parameter.



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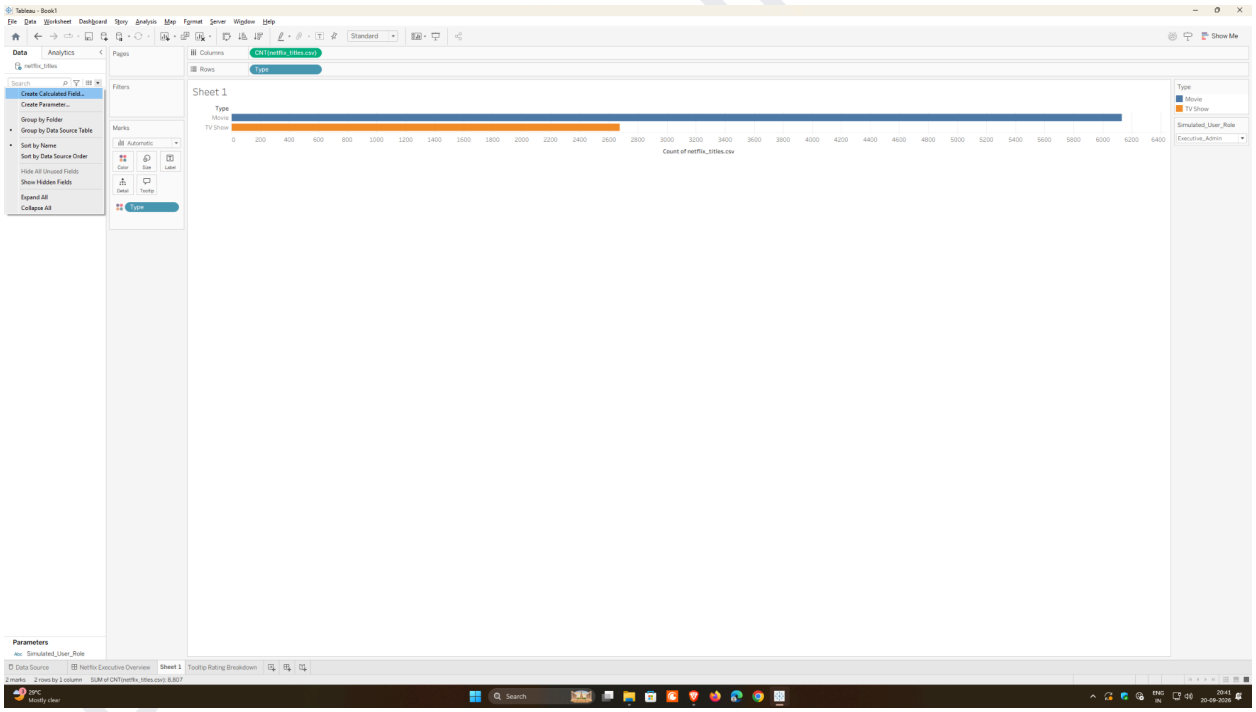
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Step 2: Write the Access Control Calculated Field

1. Right-click an empty space in the Data Pane --> select Create Calculated Field....
2. Configure the field:
   - Field Name: RLS_Access_Control
   - Tableau Formula:

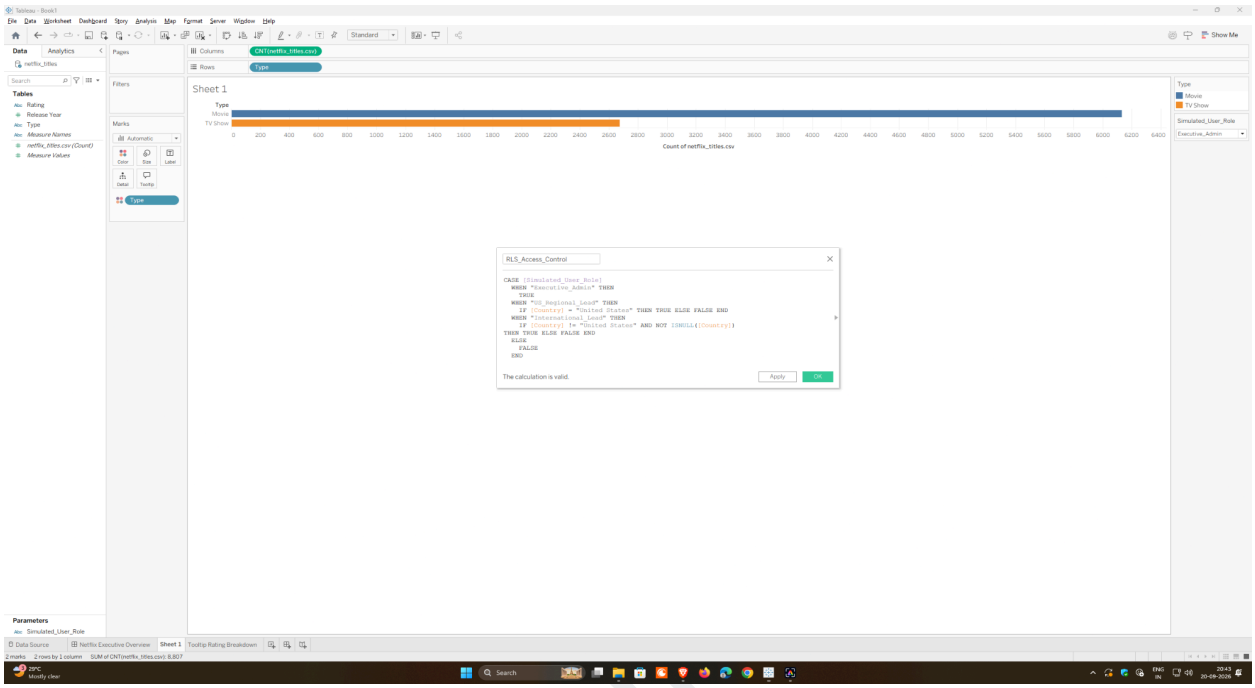
```
CASE [Simulated_User_Role]
  WHEN "Executive_Admin" THEN
    TRUE
  WHEN "US_Regional_Lead" THEN
    IF [Country] = "United States" THEN TRUE ELSE FALSE END
  WHEN "International_Lead" THEN
    IF [Country] != "United States" AND NOT ISNULL([Country]) THEN TRUE
ELSE
  FALSE
END
```



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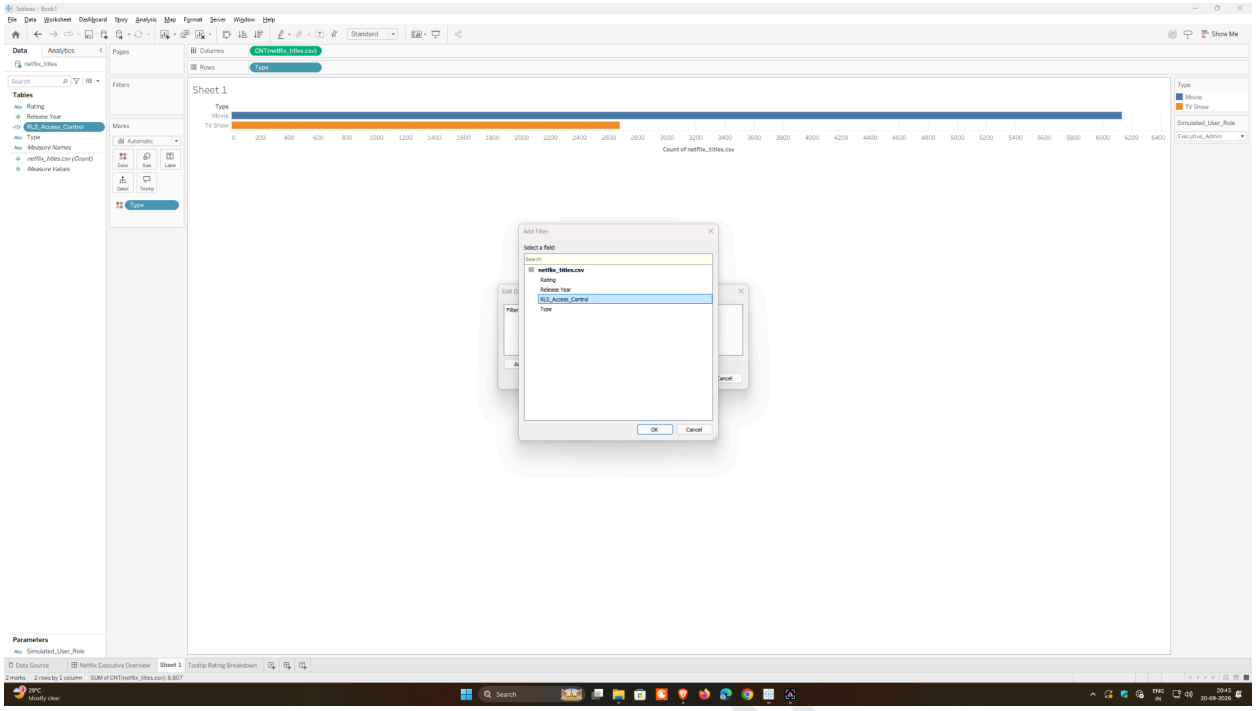
3. Click OK.

Step 3: Enforce Access Control at the Data Ingestion Layer

1. Go to the top application menu bar --> select Data --> hover over netflix_titles --> select Edit Data Source Filters....
2. In the Data Source Filters dialog, click Add....
3. Select RLS_Access_Control from the field list --> click OK.
4. In the filter dialog, check the box for True --> click OK, then click OK again.

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Step 4: Test and Verify Role-Based Permissions
Switch the Simulated_User_Role parameter dropdown on your dashboard to observe realtime access enforcement:

Select Executive_Admin: The entire global catalog of 8,807 titles across all countries is accessible.
Select US_Regional_Lead: The dashboard automatically filters to show only the 2,818 United States titles.
Select International_Lead: All US titles are masked, leaving only non-US international content.
Select Restricted_Guest: The access filter returns FALSE for all rows, blanking out the view entirely (zero data exposure).

G: Evaluate and Optimize Sharing Strategies
Step 1: Comparative Evaluation of Local Distribution Strategies

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Distribution Method	Output File Format	Recipient Requirements	Interactivity Level	Best Use Case	Key Vulnerability / Tradeoff
Packaged Workbook	.twb x	Tableau Desktop or free Tableau Reader	High (Full crossfiltering , tooltips, parameters)	Sending interactive dashboards to colleagues for standalone exploration	Embeds the entire dataset inside the file; security risk if underlying data is sensitive
Standard Workbook	.twb	Tableau Desktop + Access to original data path	High (Full authoring and filtering)	Developer version control and local script repositories	Contains only XML metadata and visualization code; will show broken links if the CSV moves
Vector Document	.pdf	Any PDF reader (Adobe, Web Browser)	None (Static printout)	Formal board packs, audited regulatory submissions , executive briefings	Cannot drill down or interact; multi-page scaling must be managed manually
High-Res Graphic	.png / .emf	Slide software (PowerPoint , Keynote)	None (Static snapshot)	Slide decks, email summaries , internal wiki articles	Fixed resolution; tooltip details and hidden marks are lost

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